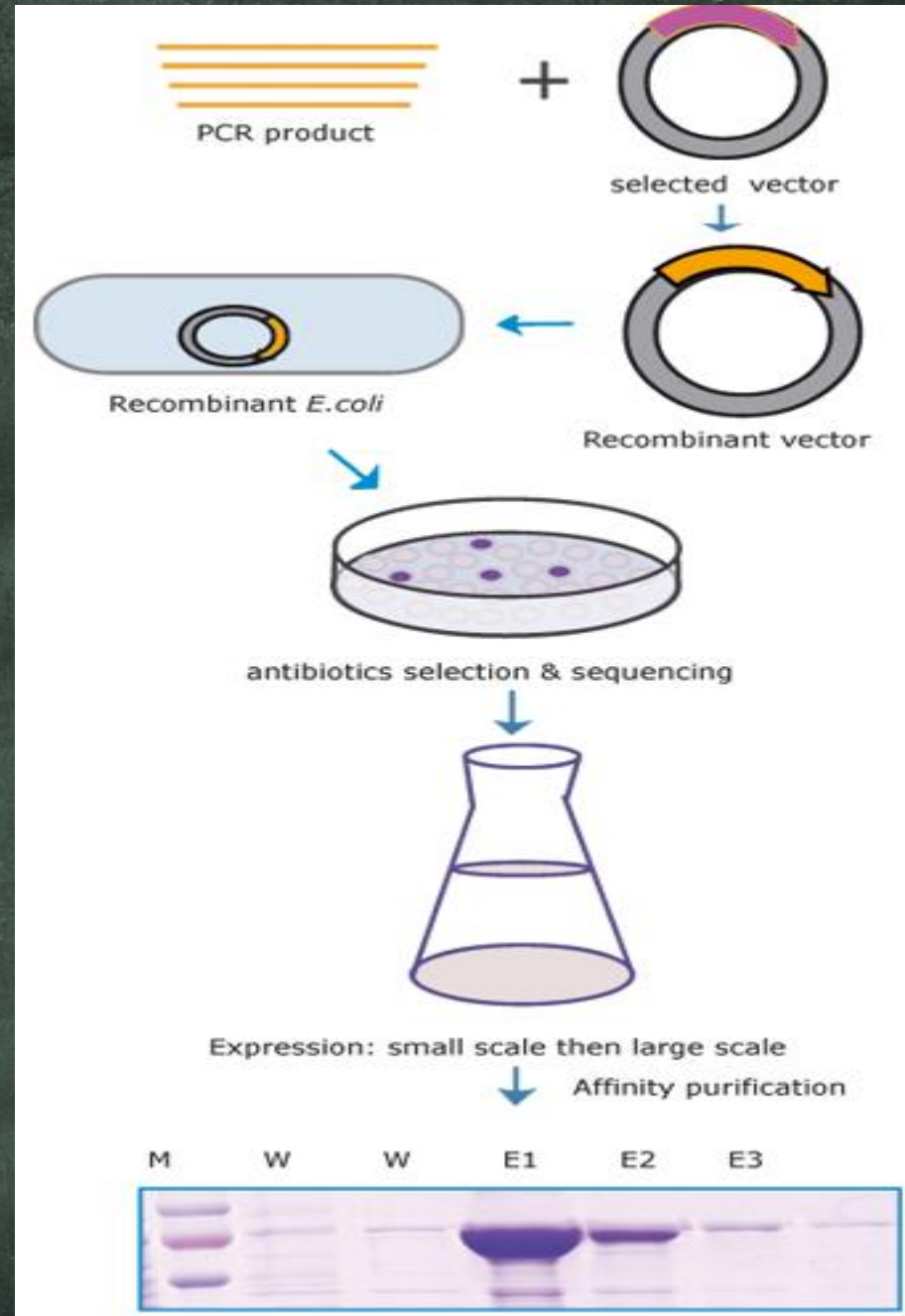
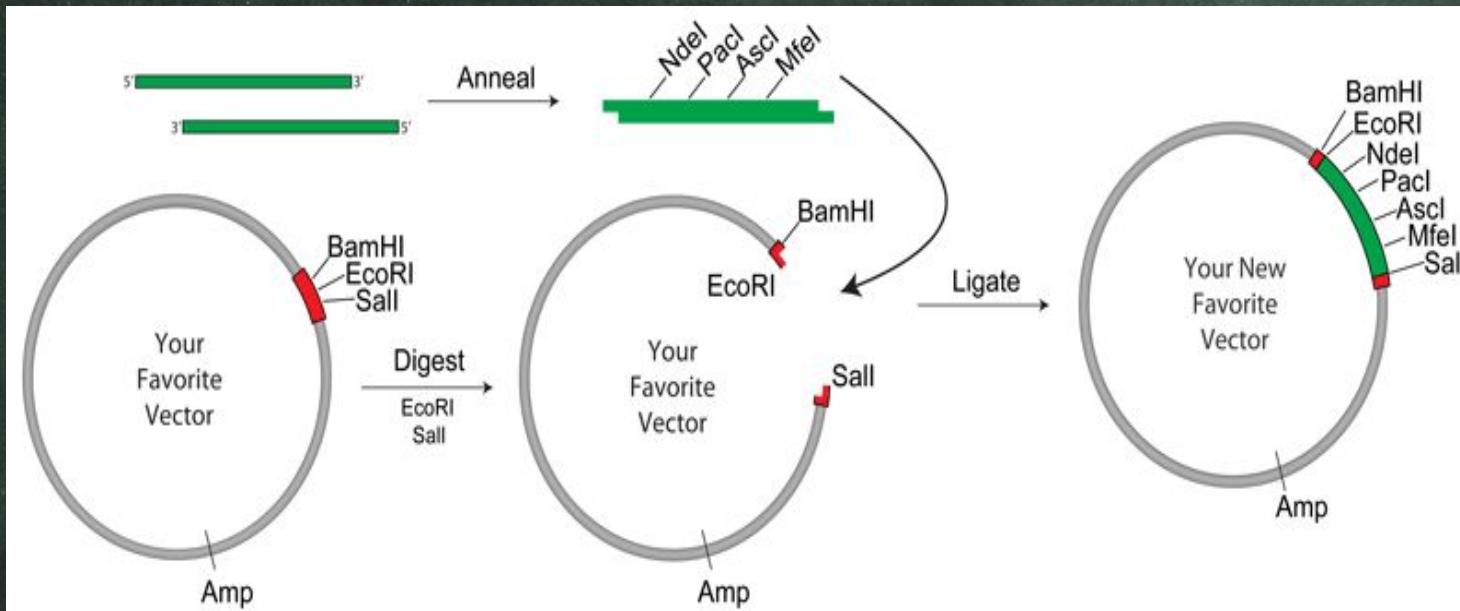




Chapter 7: SOME BASIC TECHNIQUES

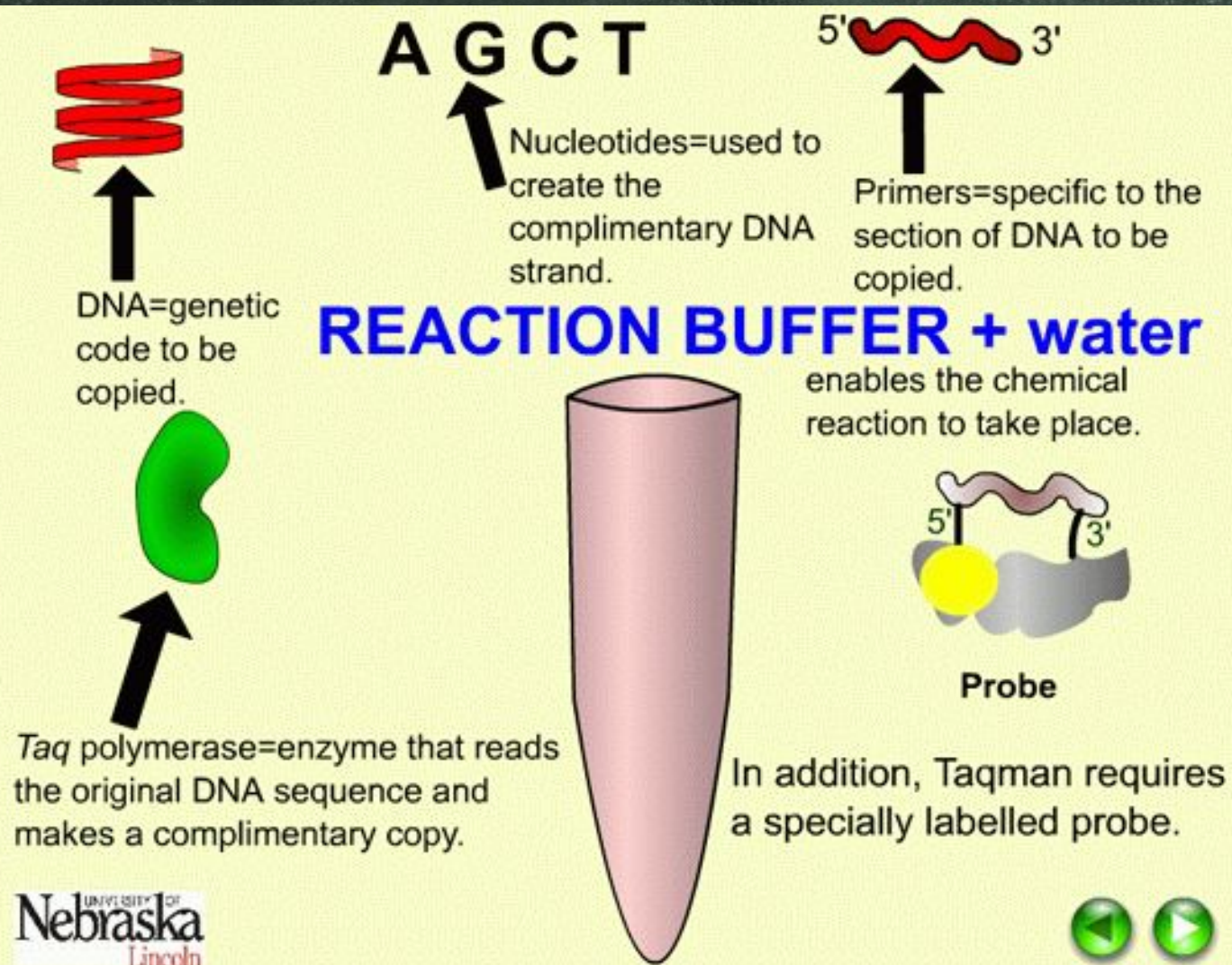


CLONING





PCR

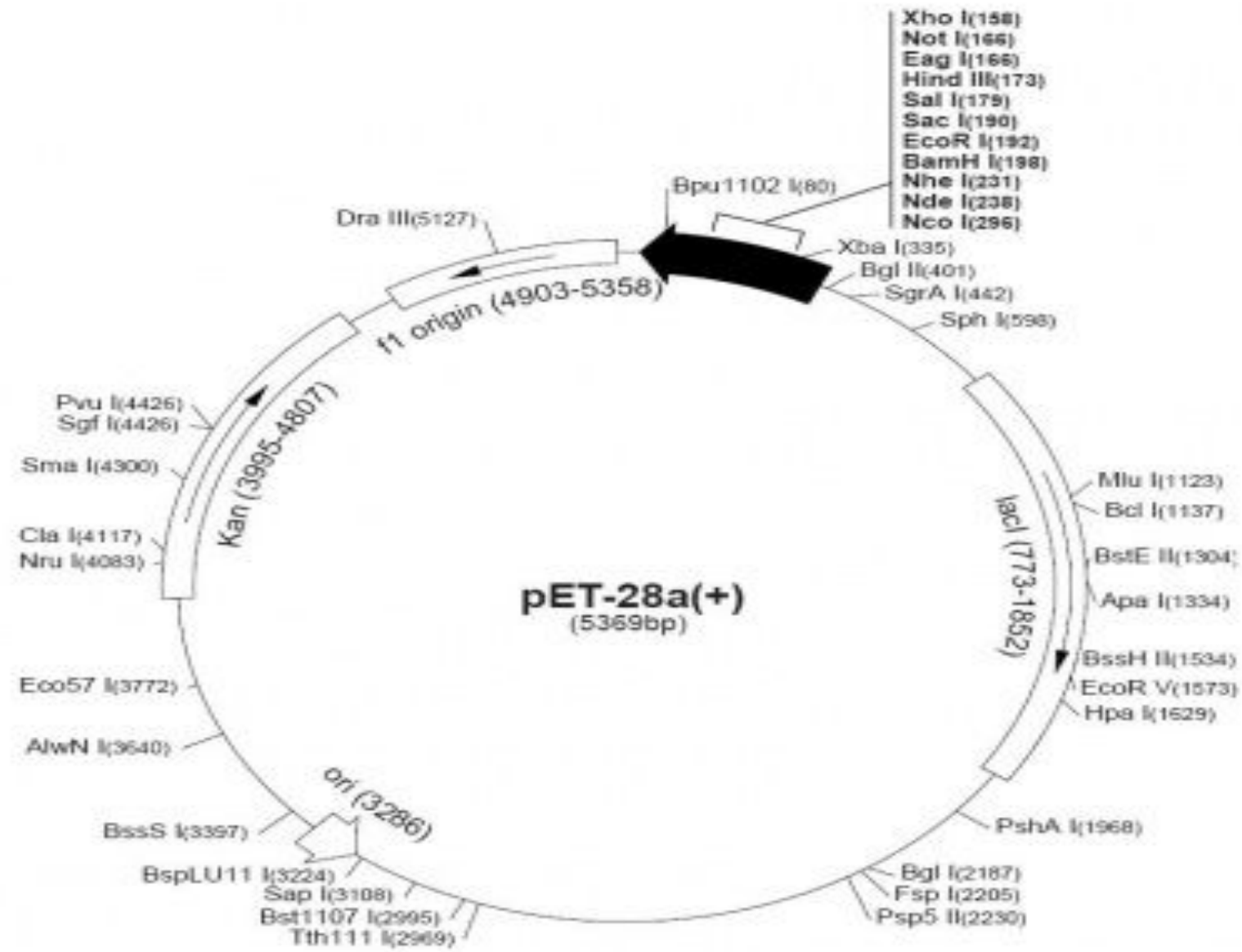




PCR routine

Thermocycling conditions for a routine PCR:

STEP	TEMP	TIME
Initial Denaturation	95°C	30 seconds
30 Cycles	95°C 45-68°C 68°C	15-30 seconds 15-60 seconds 1 minute/kb
Final Extension	68°C	5 minutes
Hold	4-10°C	



BamHI

5'... GGATCC ... 3'
3'... CCTAGG ... 5'

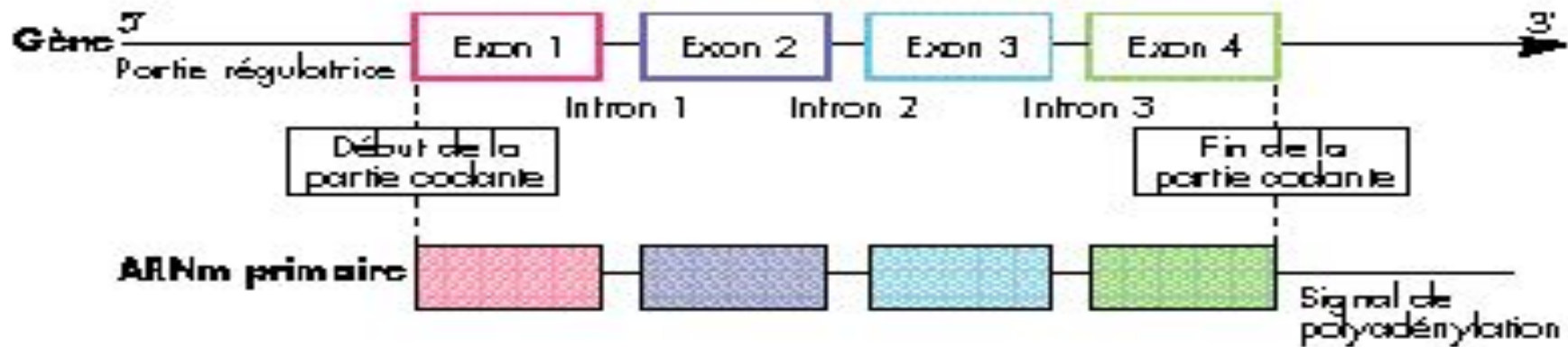
HindIII

5'... AAGCTT ... 3'
3'... TTCGAA ... 5'

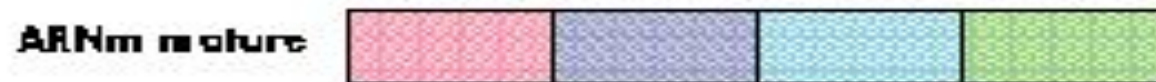


MOLECULAR HYBRIDIZATION

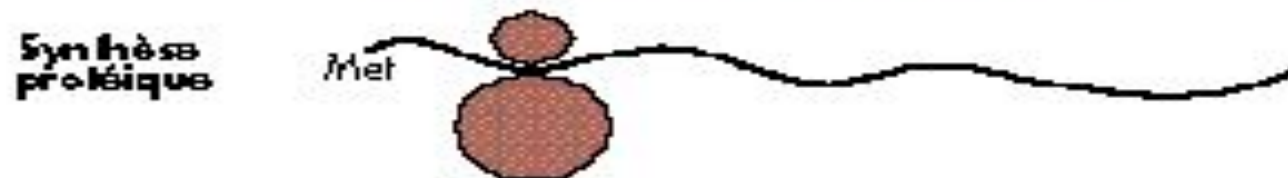
SOUTHERN BLOT



NORTHERN BLOT



WESTERN BLOT





Chapter 8 – Classical Medelian Genetics



Traits controlled by a single gene

- *Complete Dominance*

P: Yellow × Green F1: Yellow F2: 3 Yellow : 1 Green

- *Incomplete Dominance (Intermediate Inheritance)*

P: Red × White F1: Pink F2: 1 Red: 2 Pink: 1 White

- *Codominance*: Blood types (specifically the AB blood type, where both A and B antigens are expressed equally).
- *Pleiotropy*: A single gene influences multiple, seemingly unrelated phenotypic traits.



Polygenic Inheritance (Non-allelic Gene Interaction)

- **Complementary Interaction (9:7 Ratio)**

Non-allelic genes must both be present in the genotype to produce a specific phenotype.

F2: 9:7 A_B_ : Red
 A_bb, aaB_, aabb: White

- **Collaboration / New Phenotypes (9:3:3:1 Ratio)**

F2: 9:3:3:1 A_B_ : Walnut
 A_bb: Rose
 aaB_ : Pea
 aabb: Single

- **Additive Interaction / Duplicate Interaction with Cumulative Effect (9:6:1 Ratio)**

F2: 9:6:1 A_B_ : Flattened
 A_bb, aaB_ : Round
 aabb: Elongated



Epistatic Gene Interaction

An **epistatic gene** is one that **masks or suppresses** the expression of another gene (the **hypostatic gene**).

1. Dominant Suppression (13:3 Ratio)

VD: F2: 13:3 $_ _ I _ :$ White
 $C _ ii :$ Black

2. Dominant Epistasis (12:3:1 Ratio)

F2: 12:3:1 $R _ _ _ :$ Red
 $rrY _ :$ Yellow
 $rryy :$ White

3. Recessive Epistasis (9:3:4 Ratio)

F2: 9:3:4 $A _ B _ :$ Brownish-gray
 $A _ bb :$ Black
 $aaB _ , aabb :$ White

- *Additive effects*: The **phenotype** depends on the cumulative number of **dominant alleles** present in the genotype, regardless of which locus they belong to.



Chapter 9: Chromosome Genetics

The Nucleus: Genetic Library of the Cell

Chromatin and Condensed Chromosome Structure

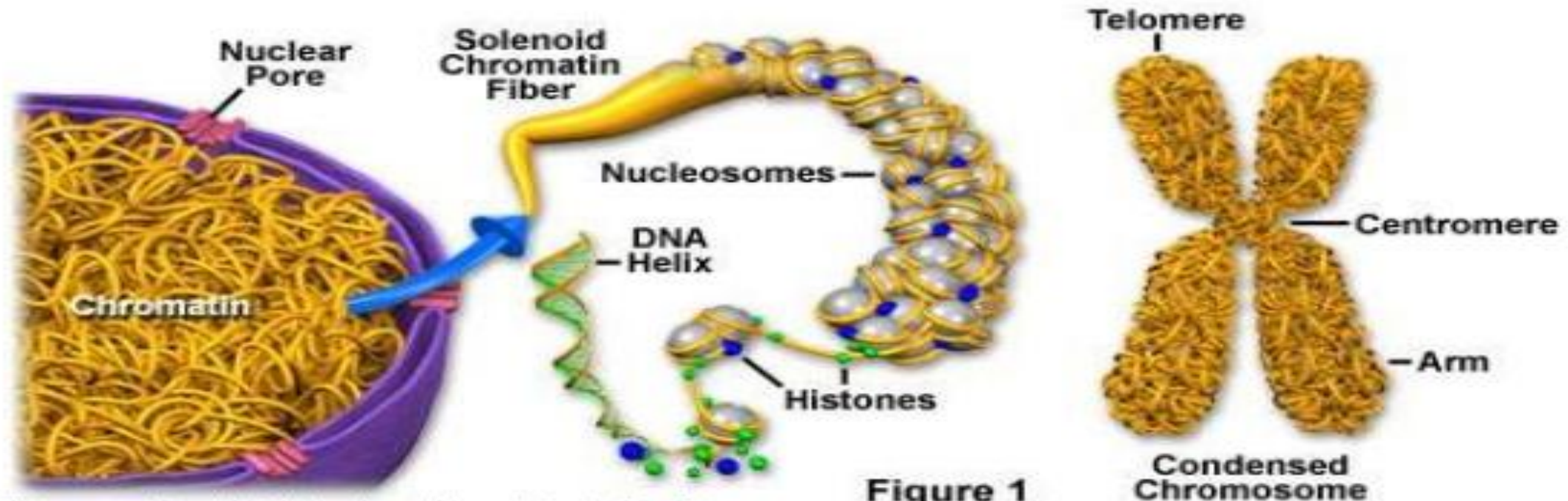
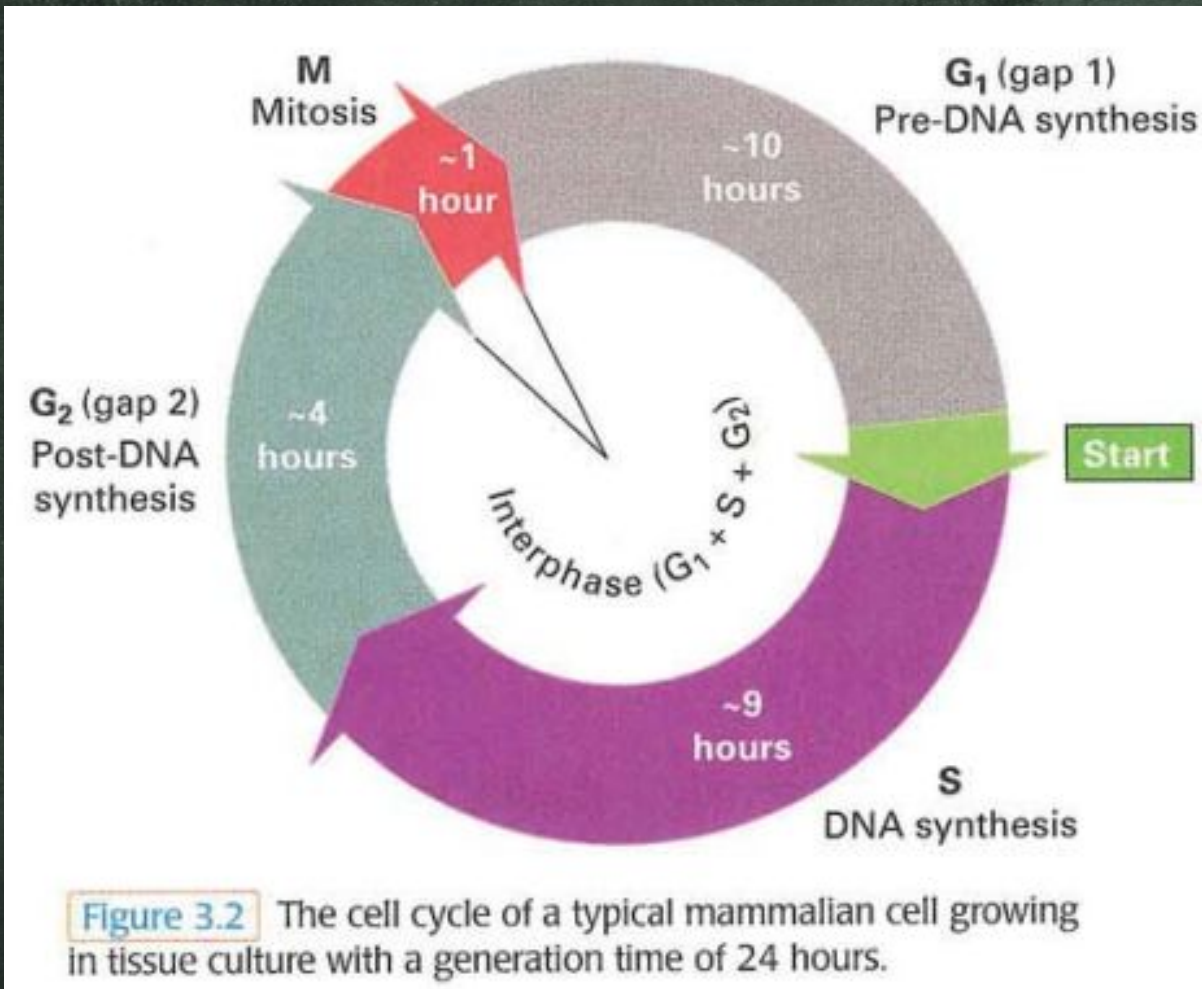


Figure 1

Condensed Chromosome

<http://micro.magnet.fsu.edu/cells/nucleus/images/chromatinstructurefigure1.jpg>

Within the nucleus the DNA is organized into discrete structures called chromosomes. Each chromosome is composed of mix of DNA and protein called chromatin. Chromatin appears diffuse in a non-dividing cell but it condenses into chromosomes when the cell prepares for cell division.



✓ G₁ phase:

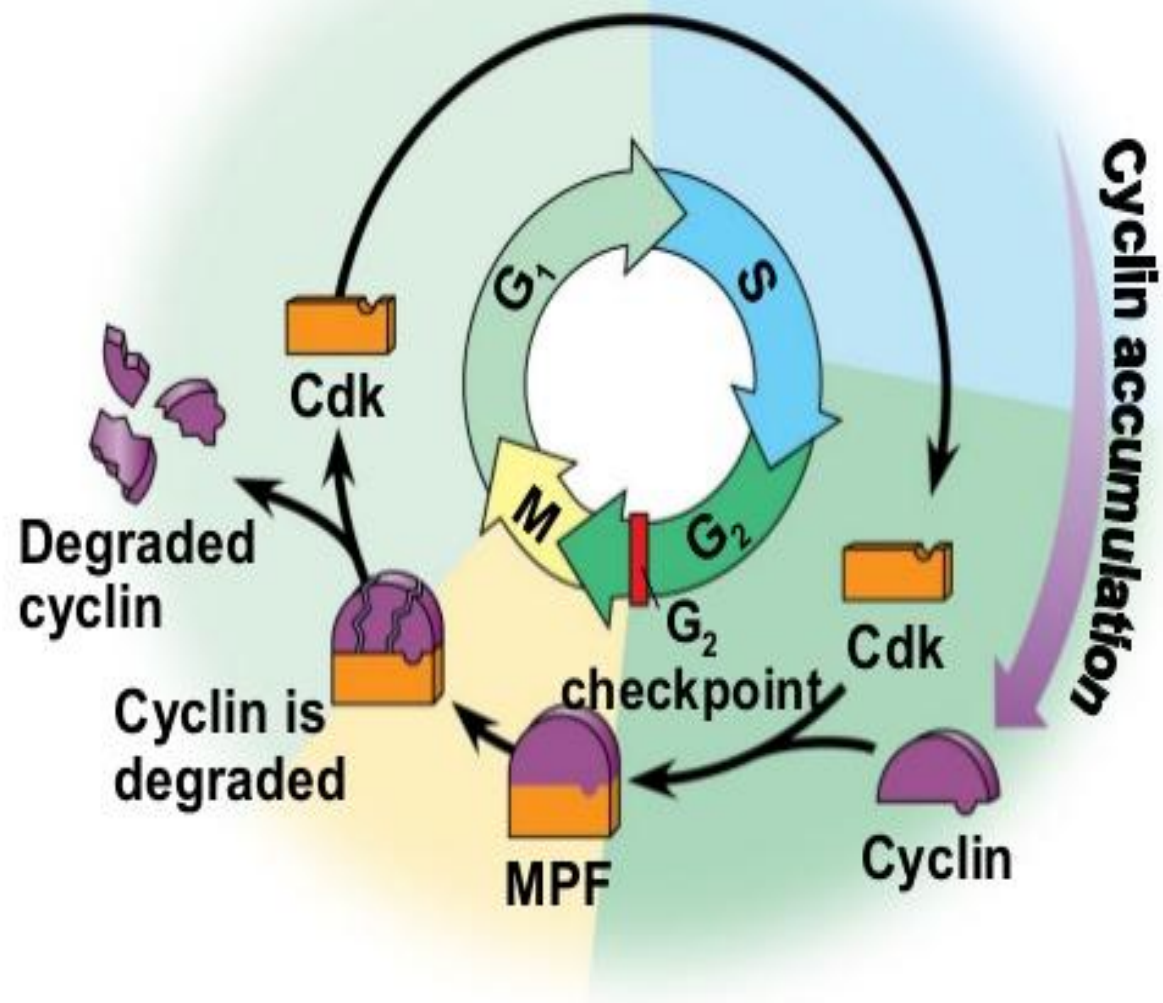
- synthesizes mRNA and proteins
- G₁/S checkpoint
- Pass to G₀ phase in limited growth factors (vertebrate)

✓ S phase:

DNA duplication

✓ G₂ phase:

- Protein synthesis for cell division
- G₂/M check point

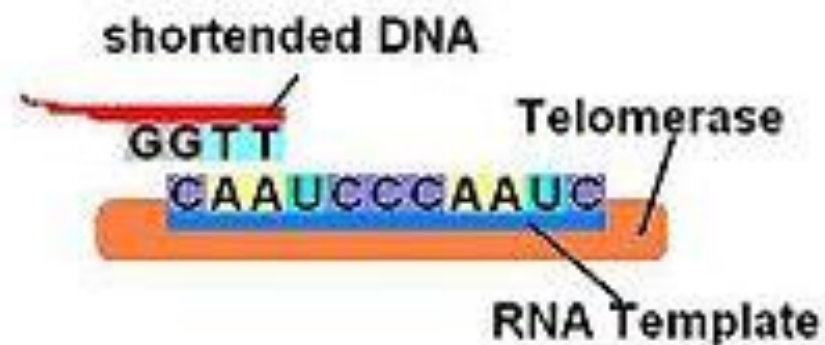
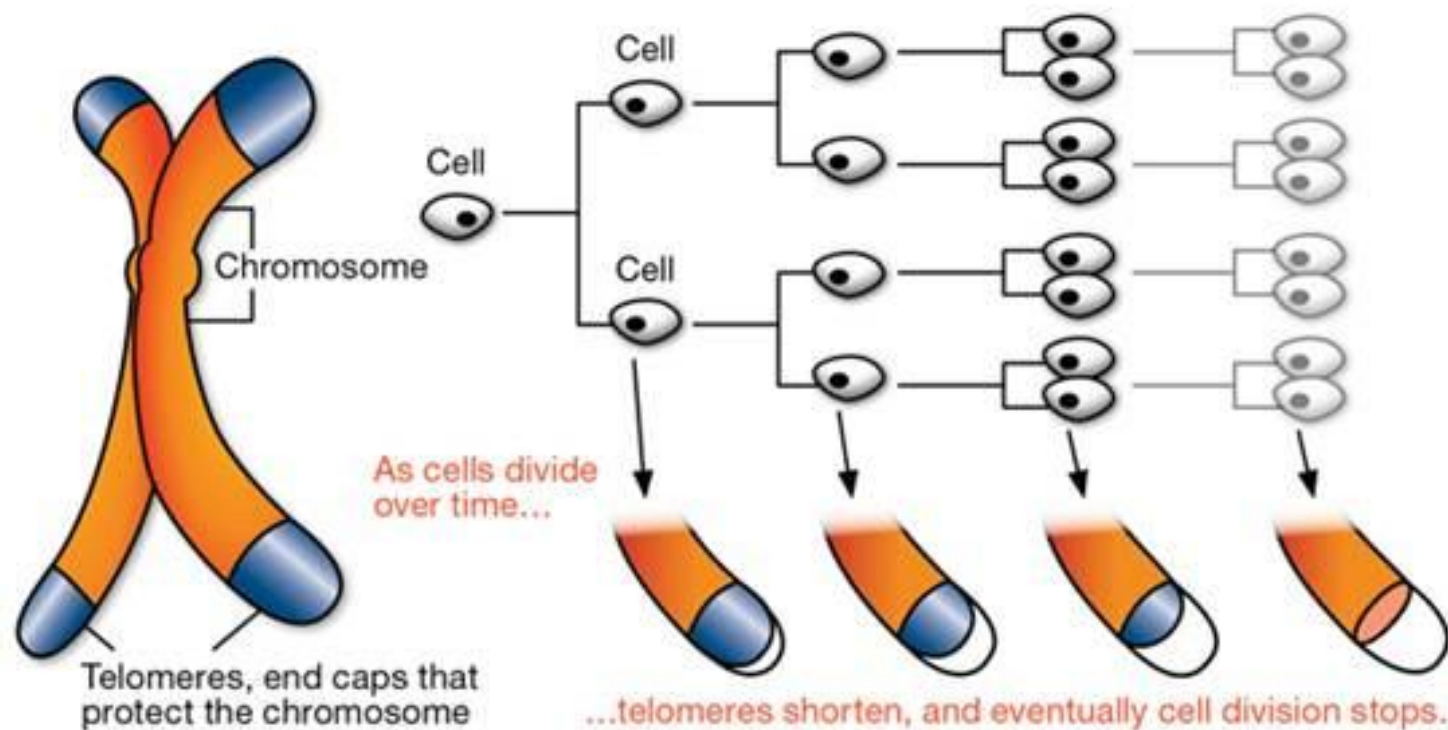




Centromere and telomeres are essential parts of chromosomes

- Centromere: essential for chromosome segregation: central component of the kinetochore □ attach & move the chromosome in both mitosis and meiosis
- Telomere: essential for the stability of the chromosome tips, elongation relies on telomerase

Telomere Shortening



DNA Polymerase attaches complementary strand





Linked inheritance

- Sex-linked Inheritance
- Linkage Inheritance:
 - Complete Linkage: Genes are always inherited together as a single unit.
 - Incomplete Linkage: Genes are separated by crossing over during meiosis.



Environmental Sex Determination (ESD) Theory

- Differentiation: Gonadal tissue
 - Cortex → Female germ cells
 - Medulla → Male germ cells
- **Gynandromorphism Mechanism:** A binucleated egg fertilized by two different sperm.



Extranuclear inheritance

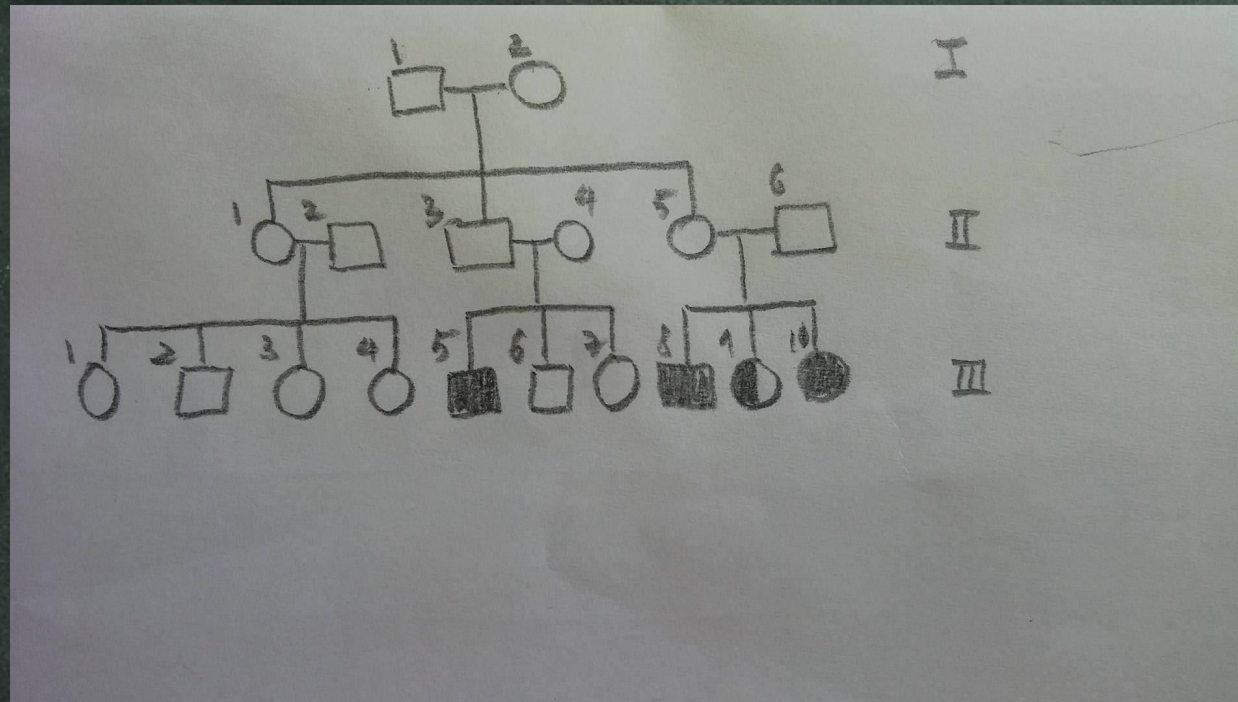
- Plastid Inheritance
- Mitochondrial Inheritance
- Cytoplasmic Male Sterility (CMS) in plants
 - Extranuclear factors determining sex/fertility versus chromosomal DNA inheritance.
 - cytS factor: Sterile cytoplasmic factor.
 - Rf factor: Nuclear restorer of fertility gene (located in the nucleus).



Complete the following pedigree chart by coloring the unidentified individuals (Generation I; Generation II-2, 3, 4, 5, 6):

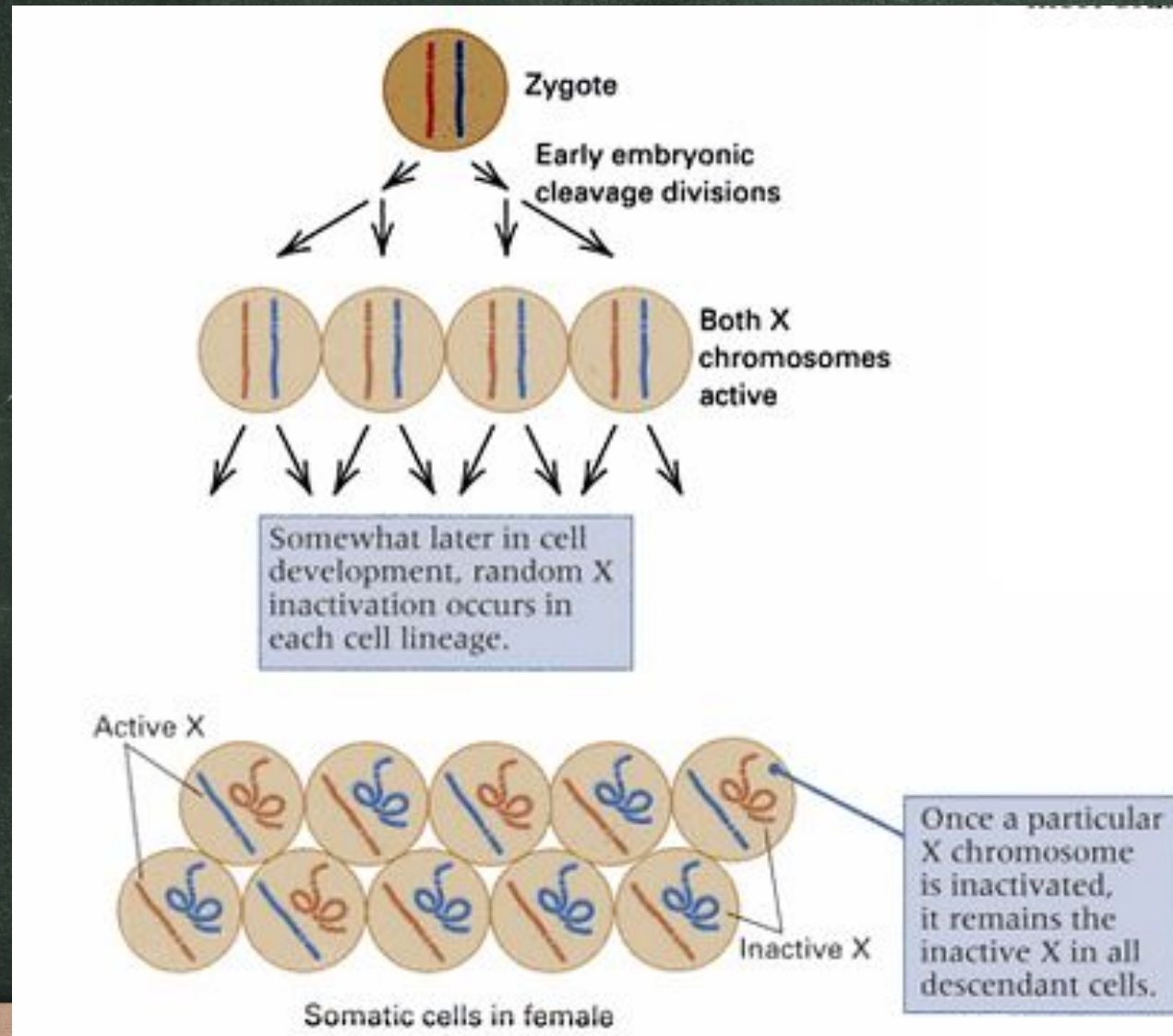
- Fully colored: Affected individual (expresses the trait).
- Half-colored: Carrier (carries the gene but does not express the trait).
- No color: Unaffected (does not carry the gene).

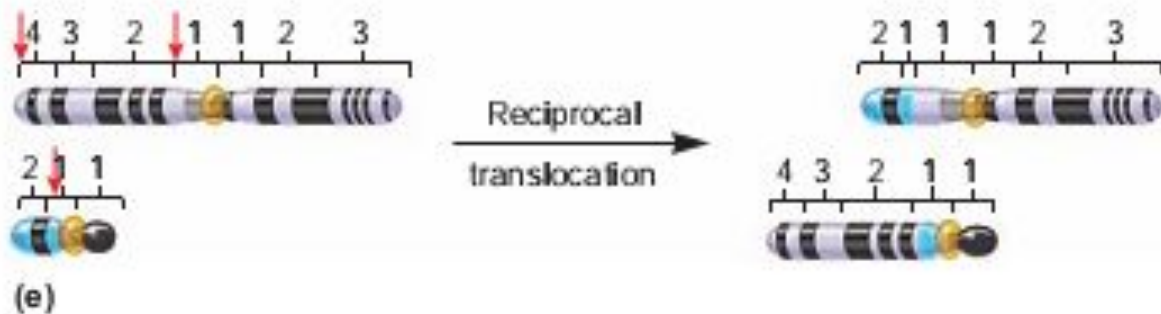
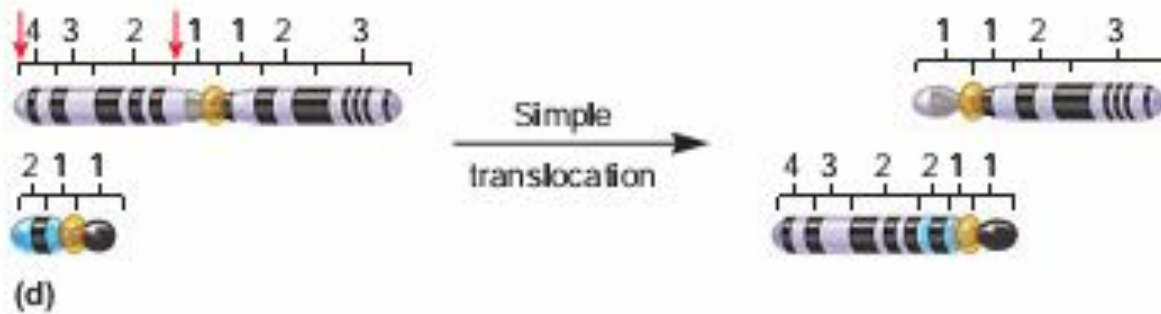
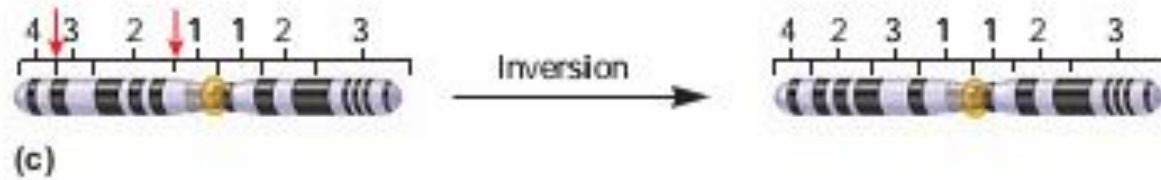
Note: Generation III and individual II-1 are already determined. The trait is sex-linked.





X-linked gene & inactive X



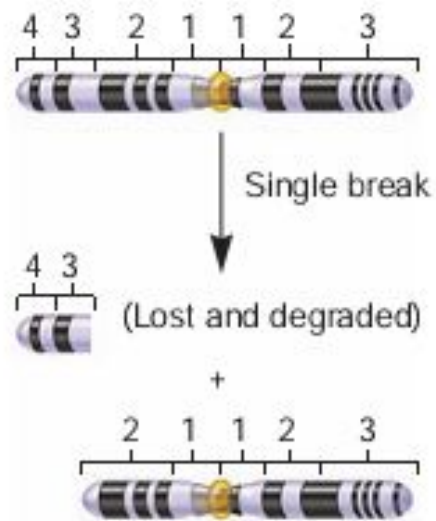


Mất đoạn

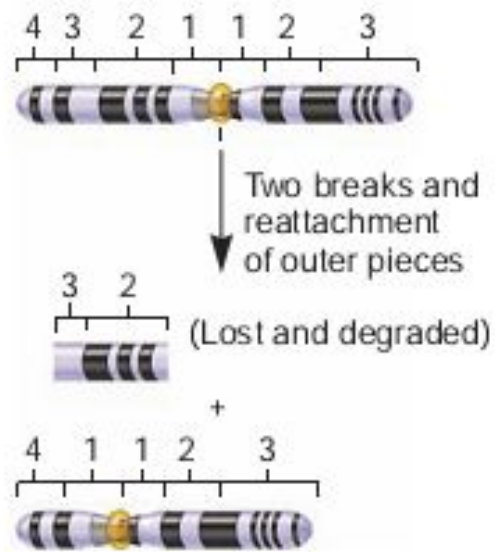
Lặp đoạn

Đảo đoạn

Chuyển đoạn



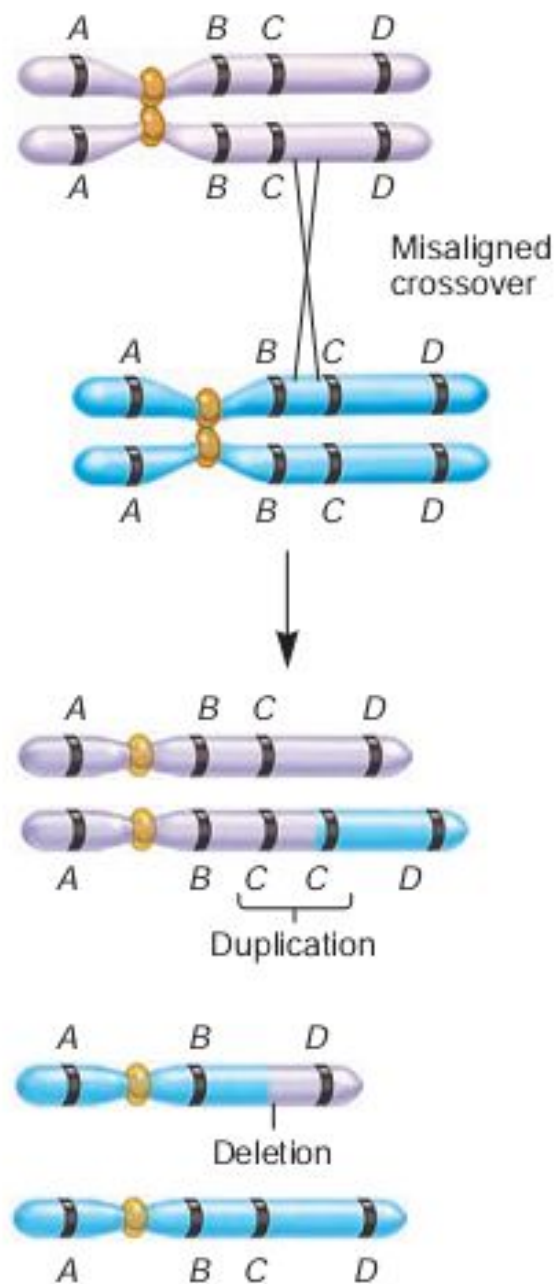
(a) Terminal deletion



(b) Interstitial deletion

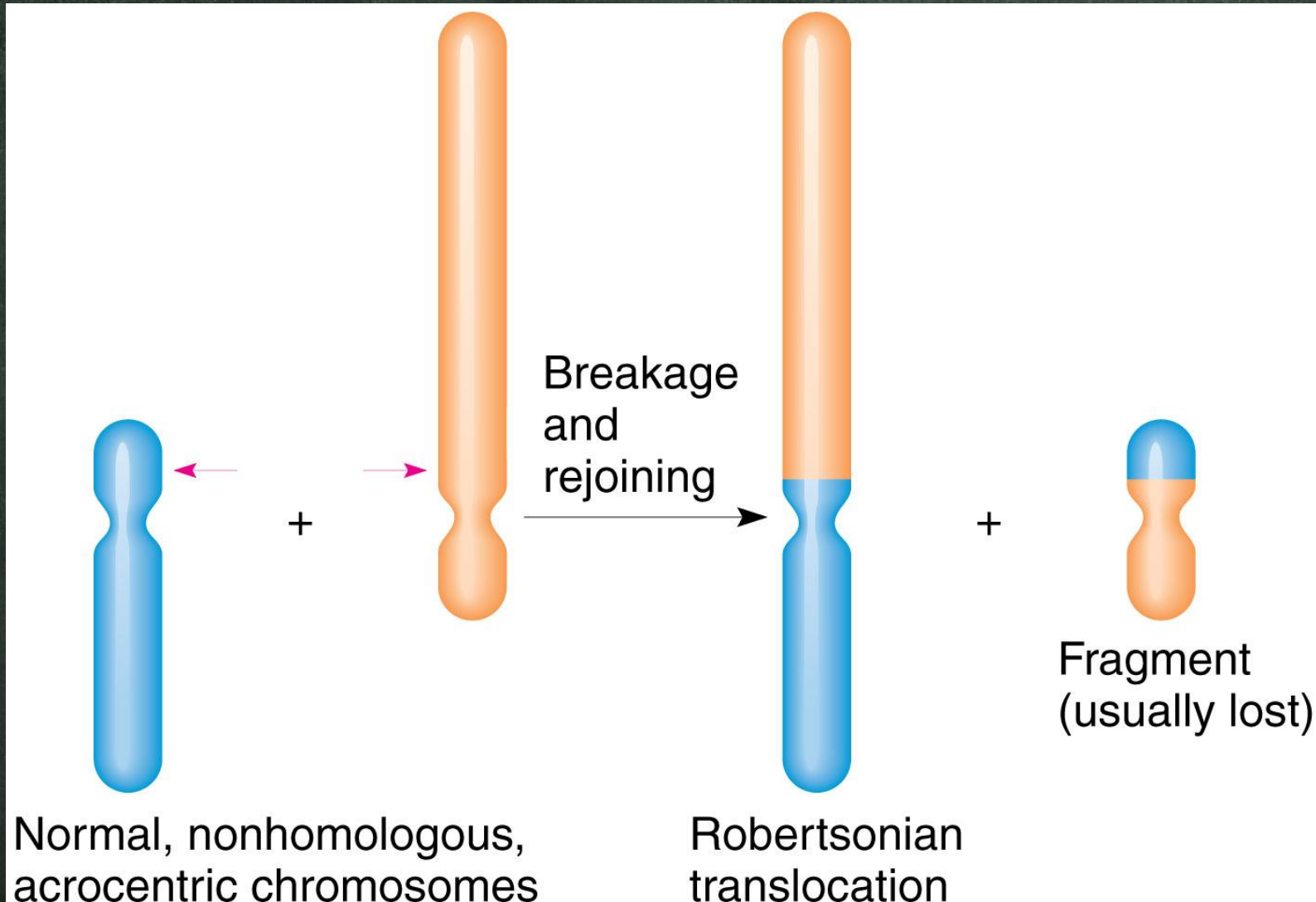
Mất đoạn đầu mút

Mất đoạn giữa

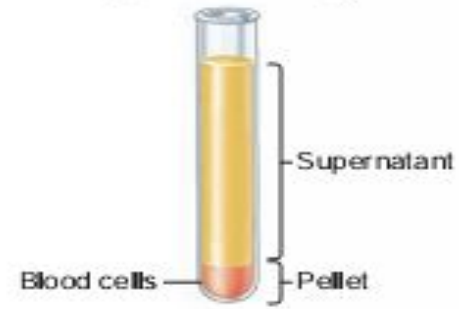




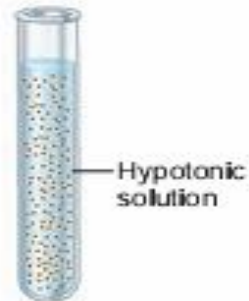
Robertsonian translocation



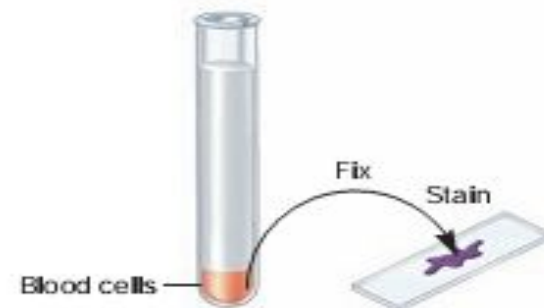
A sample of blood is collected, treated with drugs that stimulate the cells to divide, and subjected to centrifugation.



The supernatant is discarded, and the cell pellet is suspended in a hypotonic solution. This causes the cells to swell.



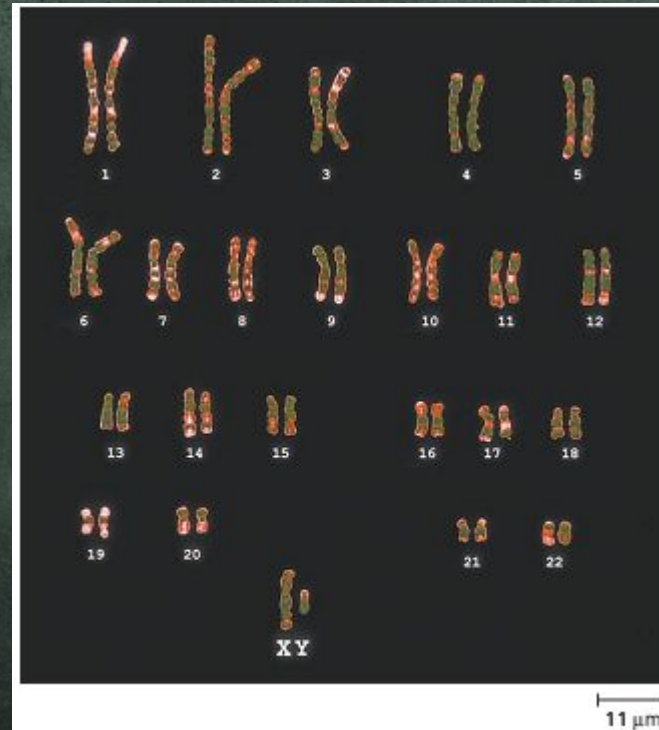
The sample is subjected to centrifugation a second time to concentrate the cells. The cells are suspended in a fixative, stained, and placed on a slide.



(a) Preparing cells for a karyotype



(b) The slide is viewed by a light microscope; the sample is seen on a video screen. The chromosomes can be arranged electronically on the screen.



(c) For a diploid human cell, two complete sets of chromosomes from a single cell constitute a karyotype of that cell.

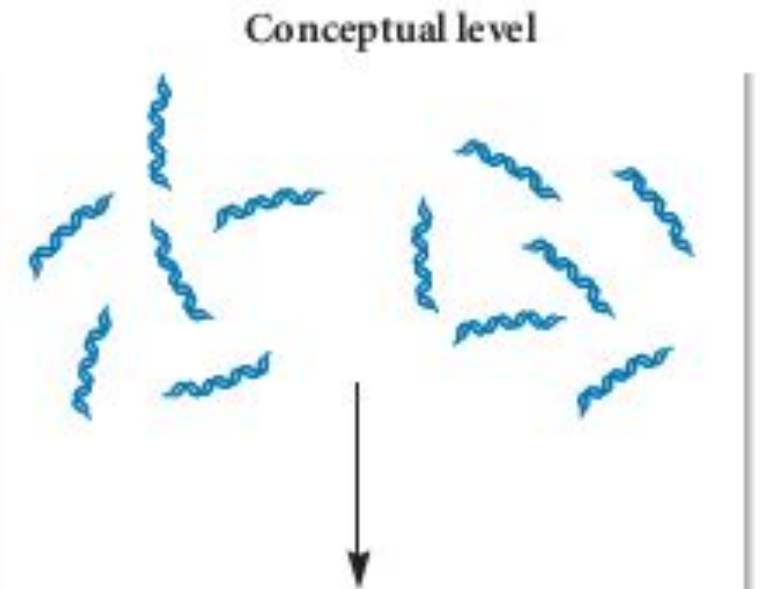
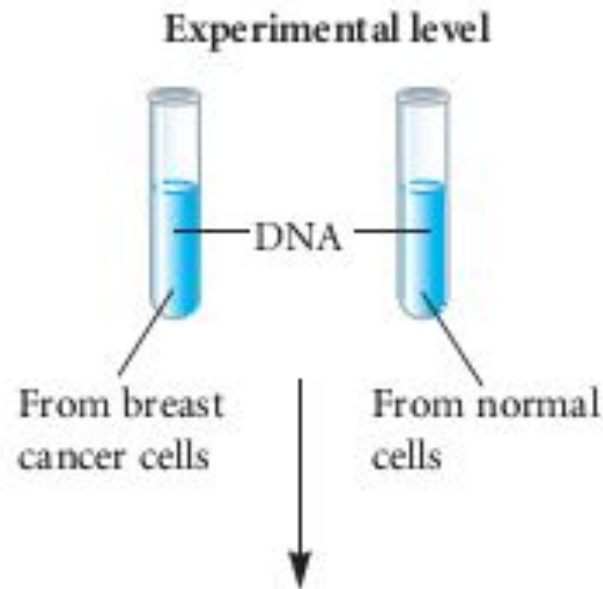


Comparative genomic hybridization

ACHIEVING THE GOAL — FIGURE 8.8 The use of comparative genomic hybridization to detect deletions and duplications in cancer cells.

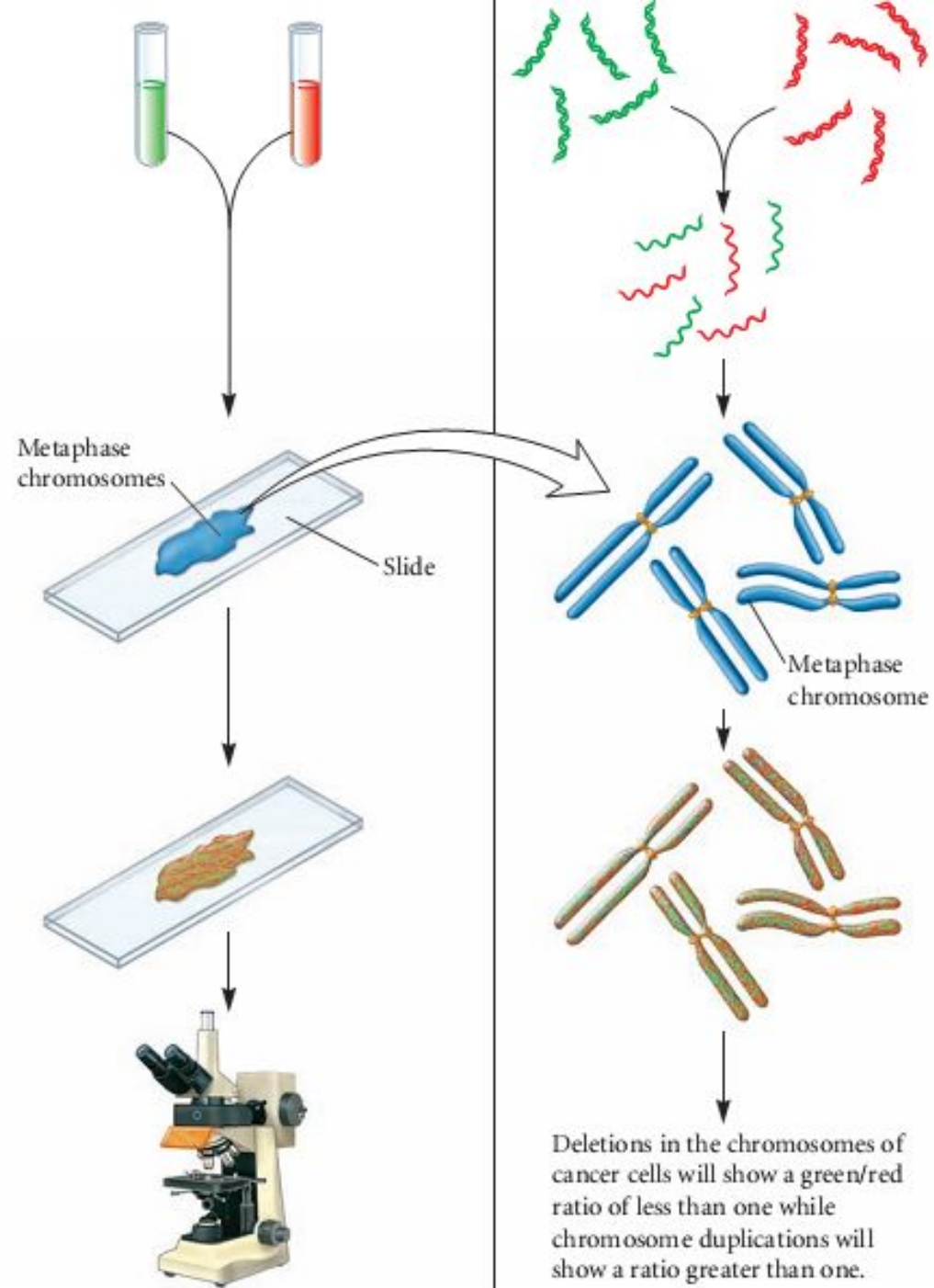
Starting materials: Breast cancer cells and normal cells.

1. Isolate DNA from human breast cancer cells and normal cells. This involved breaking open the cells and isolating the DNA by chromatography. (See Appendix for description of chromatography.)



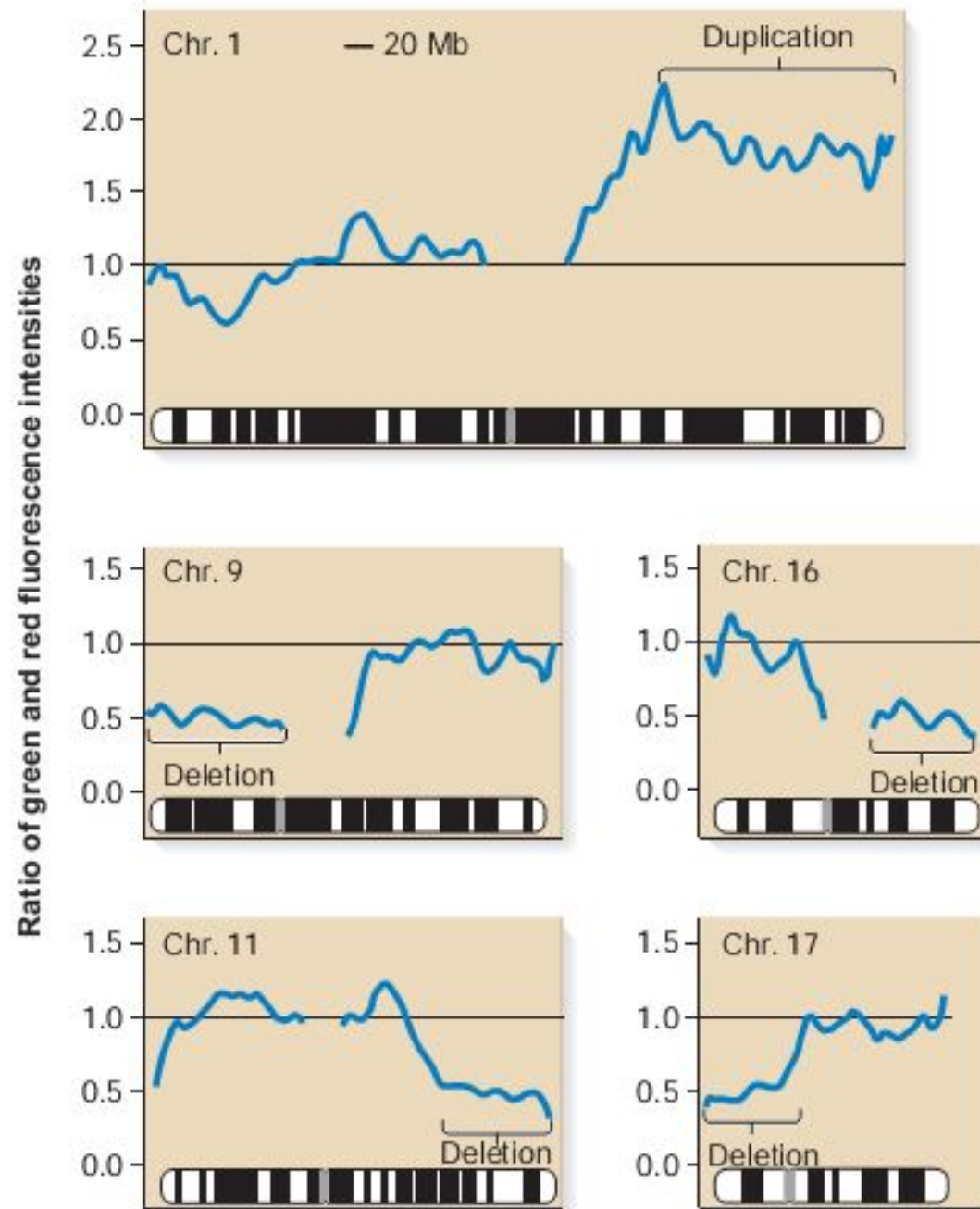


2. Label the breast cancer DNA with a green fluorescent molecule and the normal DNA with a red fluorescent molecule. This was done by using the DNA from step 1 as a template, and incorporating fluorescently labeled nucleotides into newly made DNA strands.
3. The DNA strands were then denatured by heat treatment. Mix together equal amounts of fluorescently labeled DNA and add it to a preparation of metaphase chromosomes from white blood cells. The procedure for preparing metaphase chromosomes is described in Figure 3.2. The metaphase chromosomes were also denatured.
4. Allow the fluorescently labeled DNA to hybridize to the metaphase chromosomes.
5. Visualize the chromosomes with a fluorescence microscope. Analyze the amount of green and red fluorescence along each chromosome with a computer.



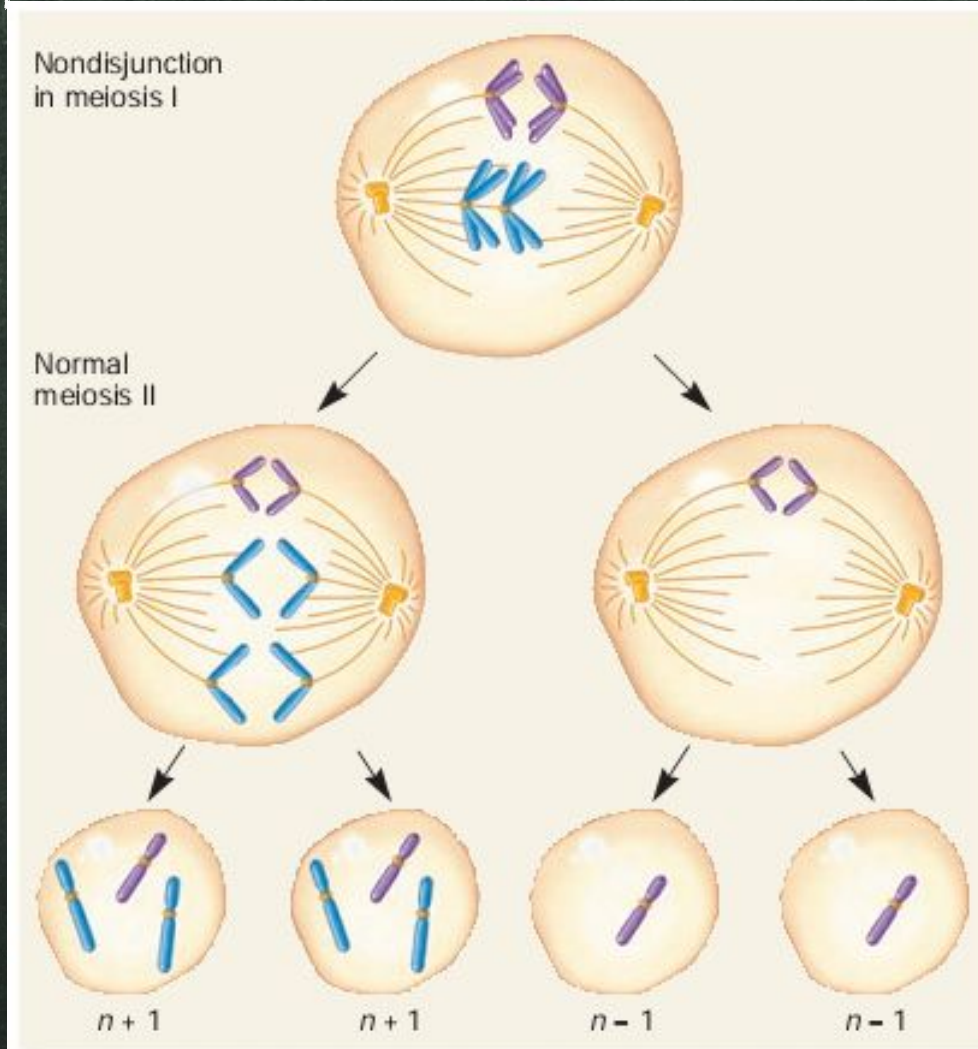


THE DATA

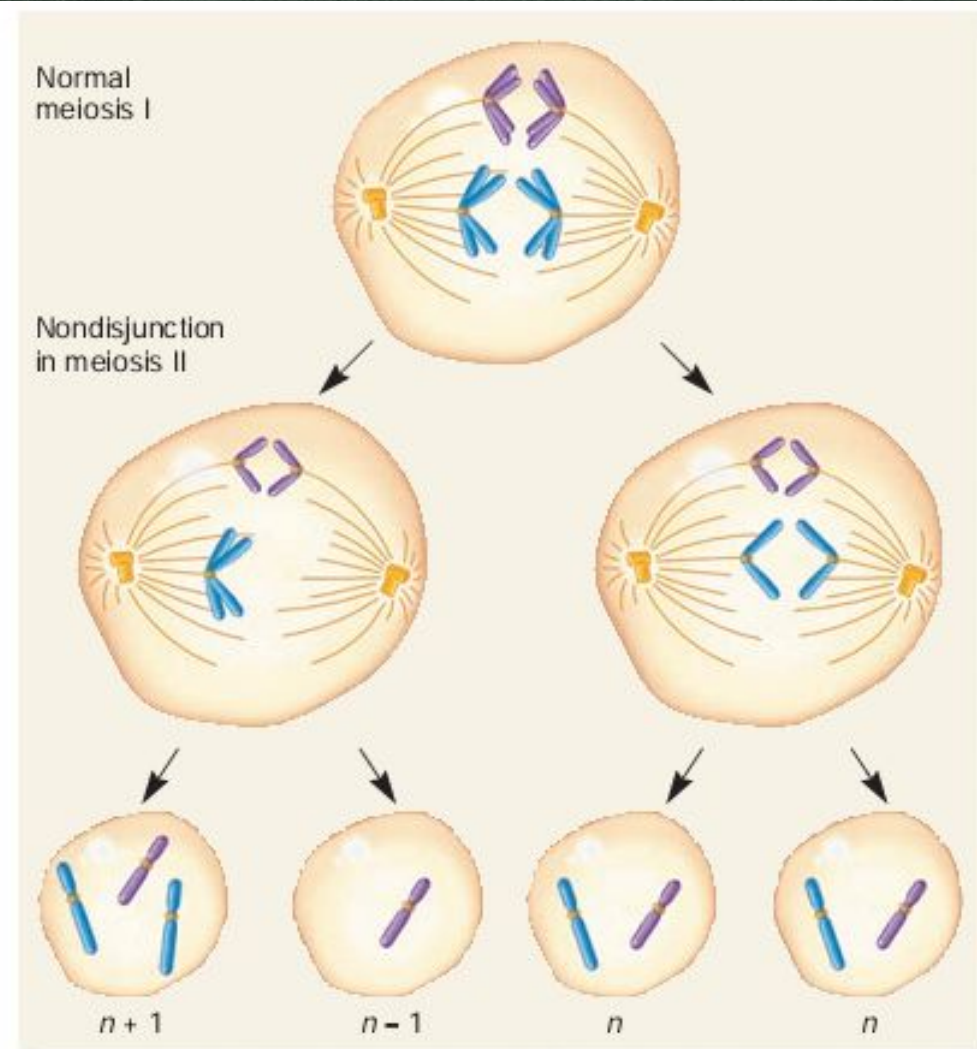




Chromosome number anomalies



(a) Nondisjunction in meiosis I

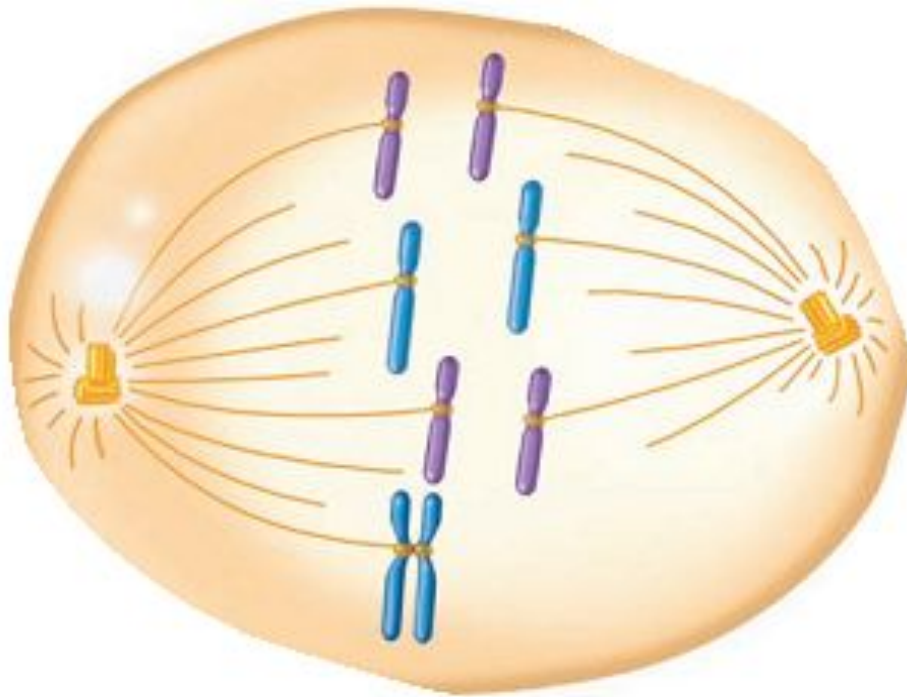


(b) Nondisjunction in meiosis II

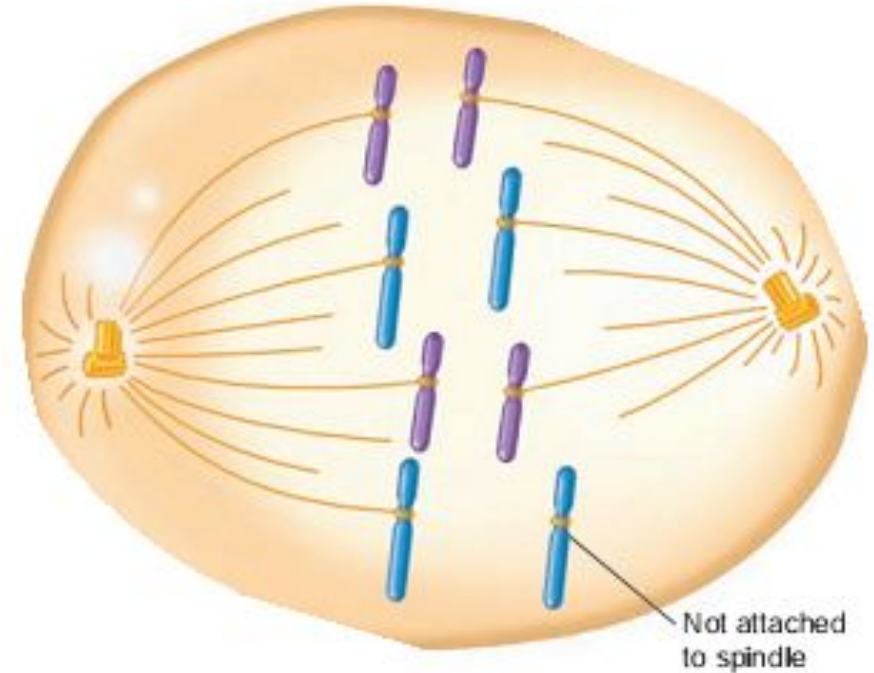


Chromosome number anomalies & diseases

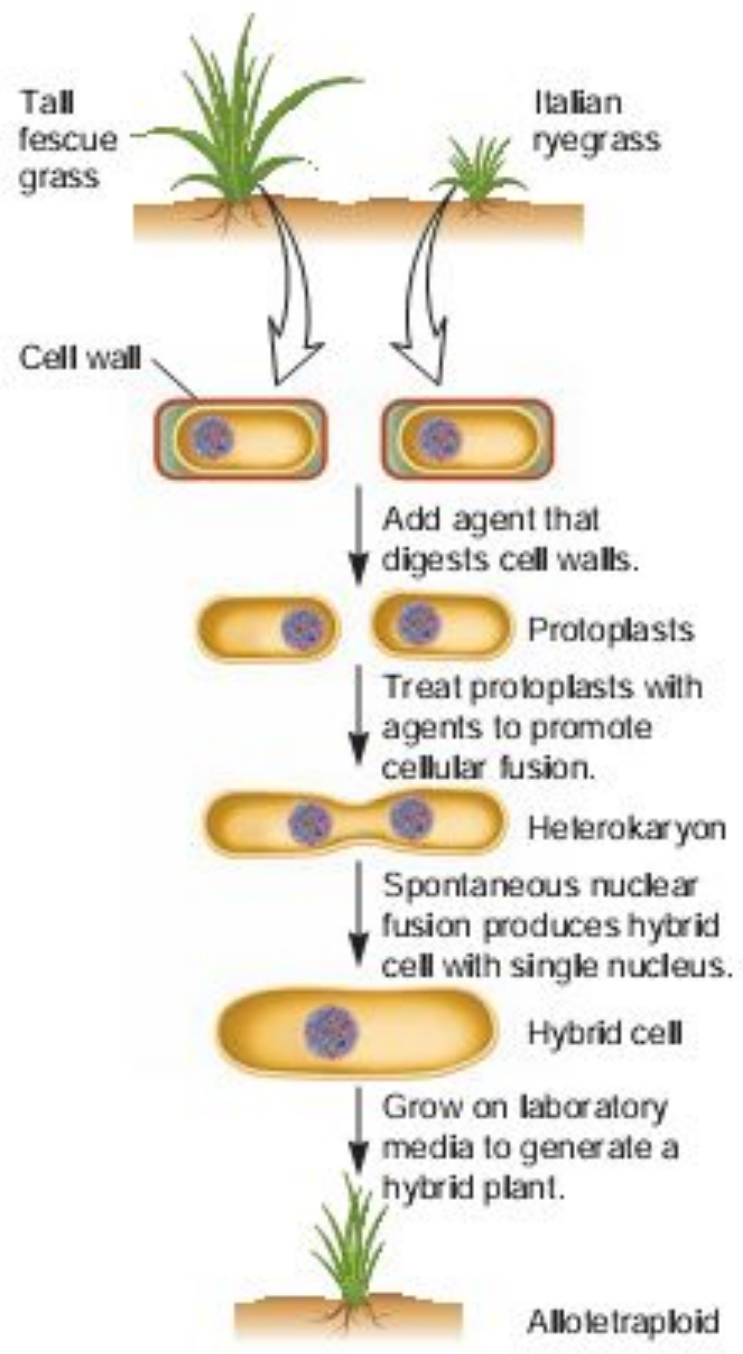
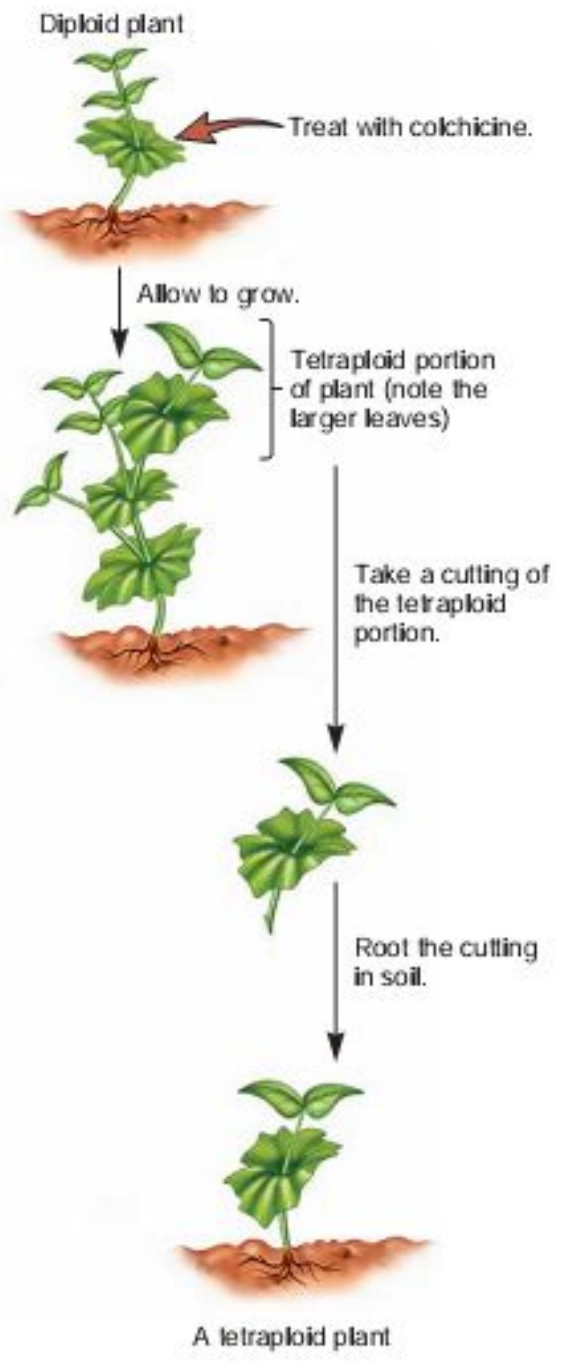
- **Down syndrome:** Trisomy 21 (3 copies of chromosome 21)
- **47, XXX:** trisomy-X syndrome
- **47, XYY:** double-Y syndrome
- **47, XXY:** Klinefelter syndrome
- **45, X:** monosomy of the X chromosome in female called Turner syndrome

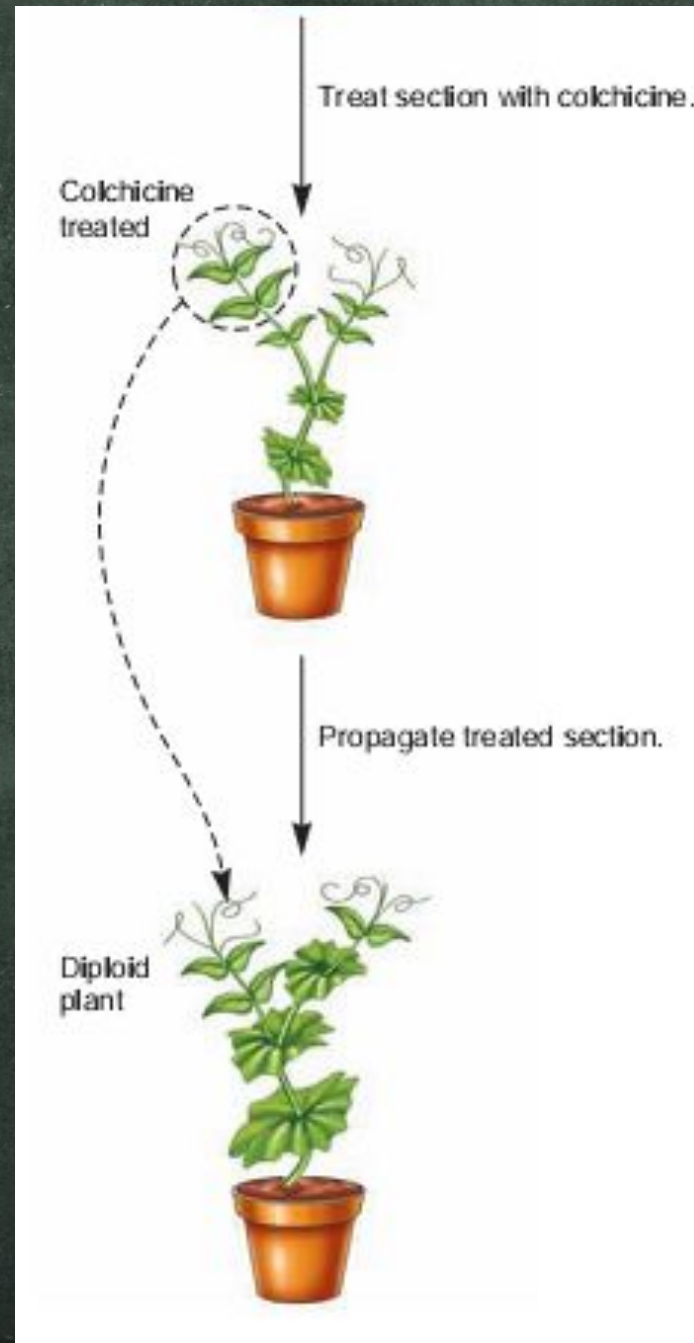
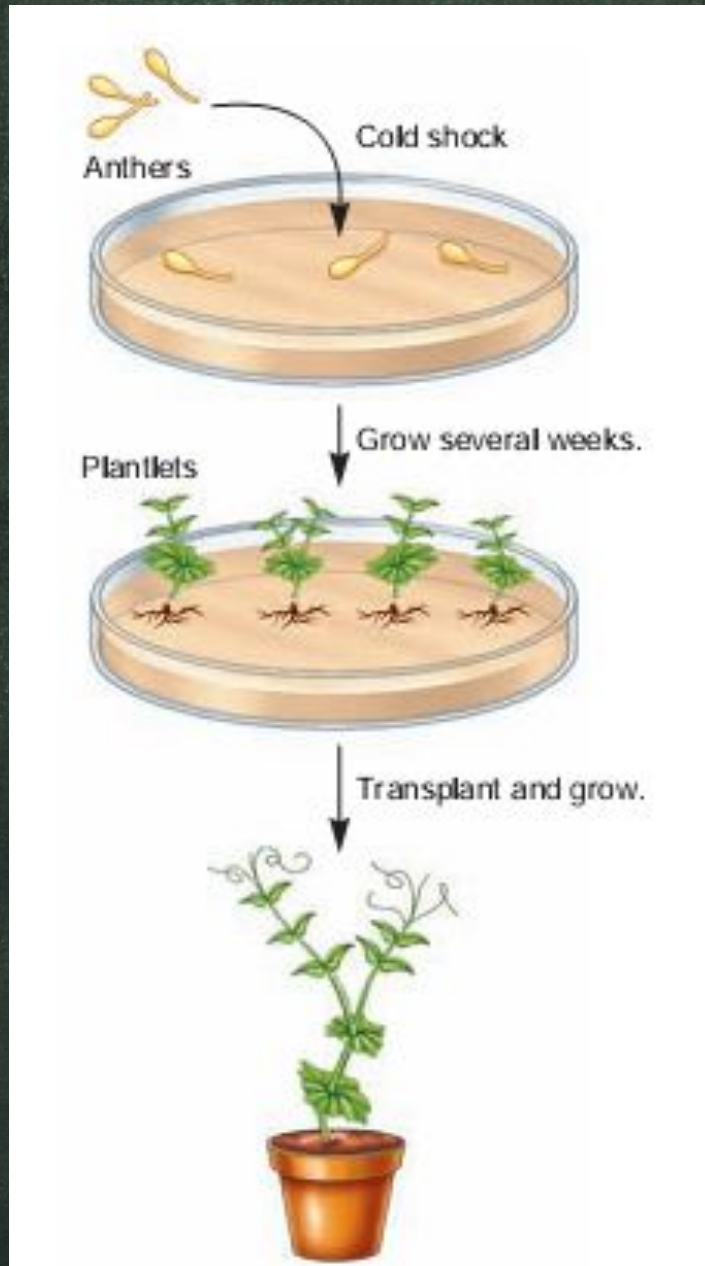


(a) Mitotic nondisjunction

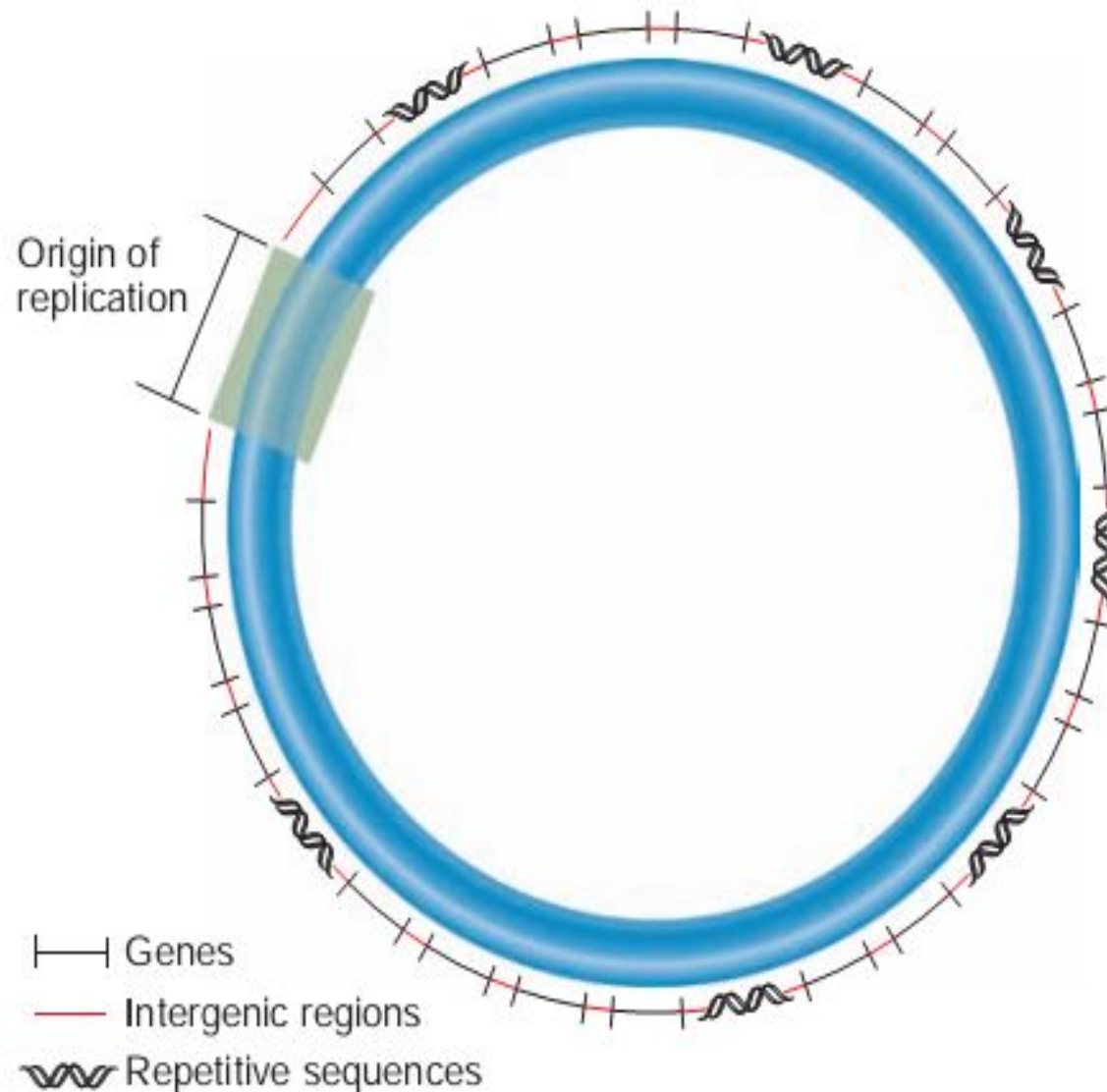


(b) Chromosome loss



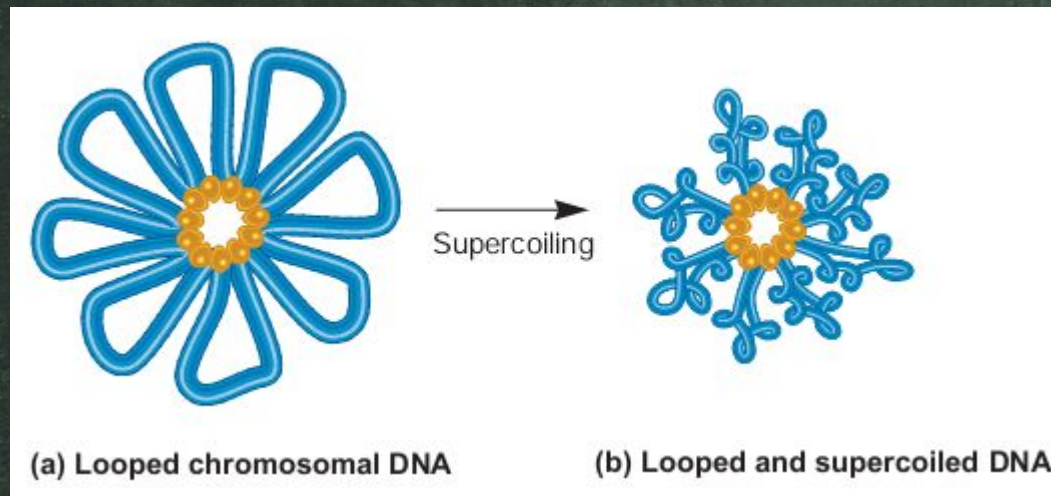
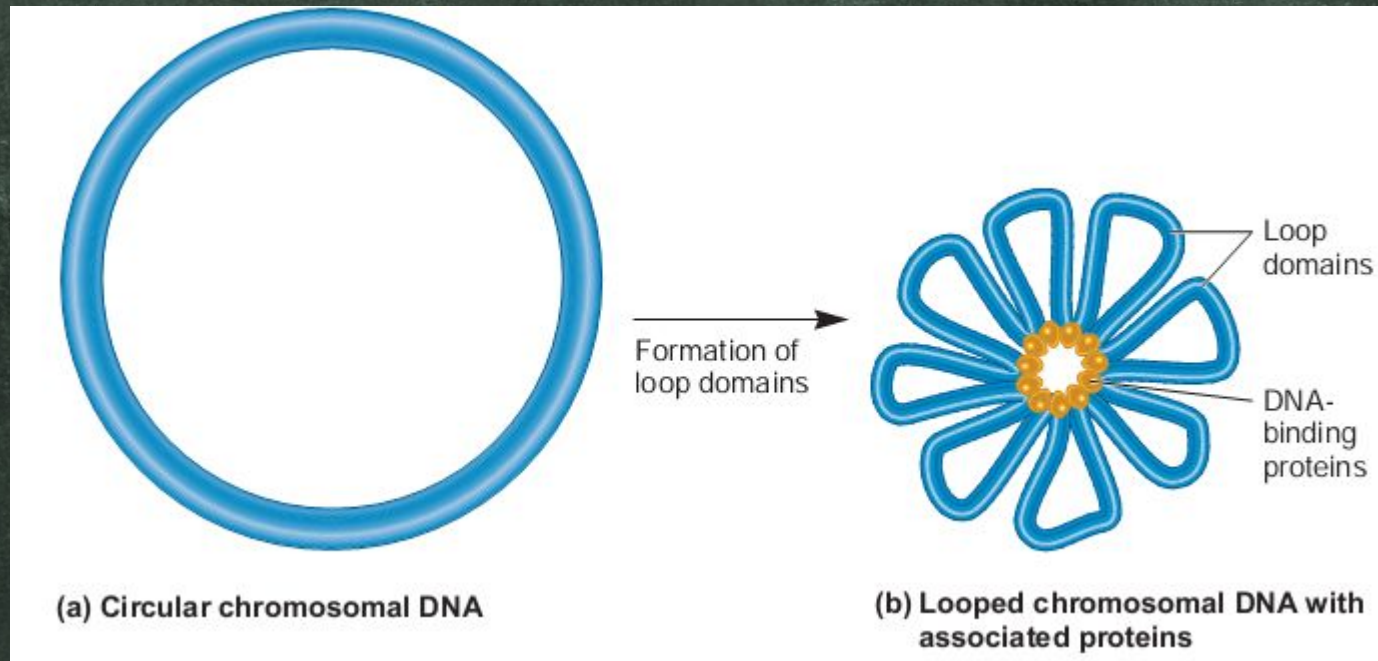


Chapter 10: Bacterial Genetics



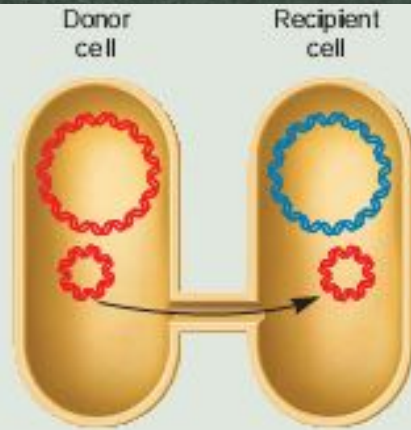
Key features:

- Most, but not all, bacterial species contain circular chromosomal DNA.
- A typical chromosome is a few million base pairs in length.
- Most bacterial species contain a single type of chromosome, but it may be present in multiple copies.
- Several thousand different genes are interspersed throughout the chromosome. The short regions between adjacent genes are called intergenic regions.
- One origin of replication is required to initiate DNA replication.
- Repetitive sequences may be interspersed throughout the chromosome.



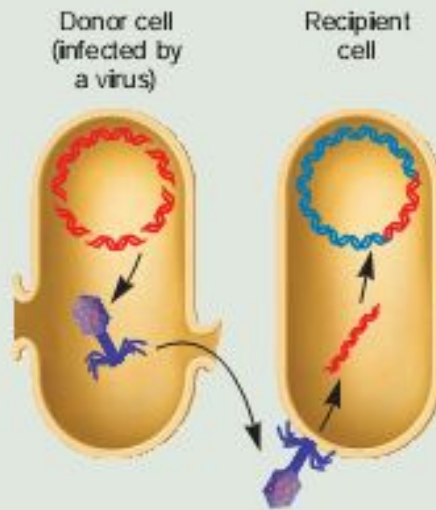


Conjugation



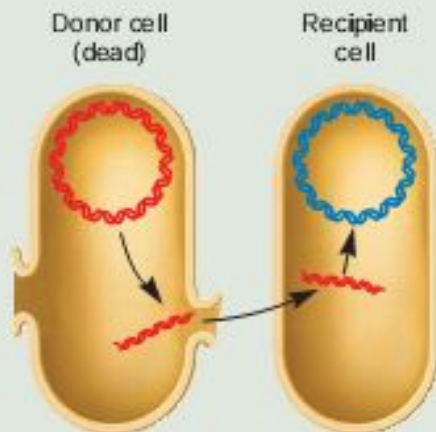
Requires direct contact between a donor and a recipient cell. The donor cell transfers a strand of DNA to the recipient. In the example shown here, DNA known as a plasmid is transferred to the recipient cell.

Transduction



When a virus infects a donor cell, it incorporates a fragment of bacterial chromosomal DNA into a newly made virus particle. The virus then transfers this fragment of DNA to a recipient cell.

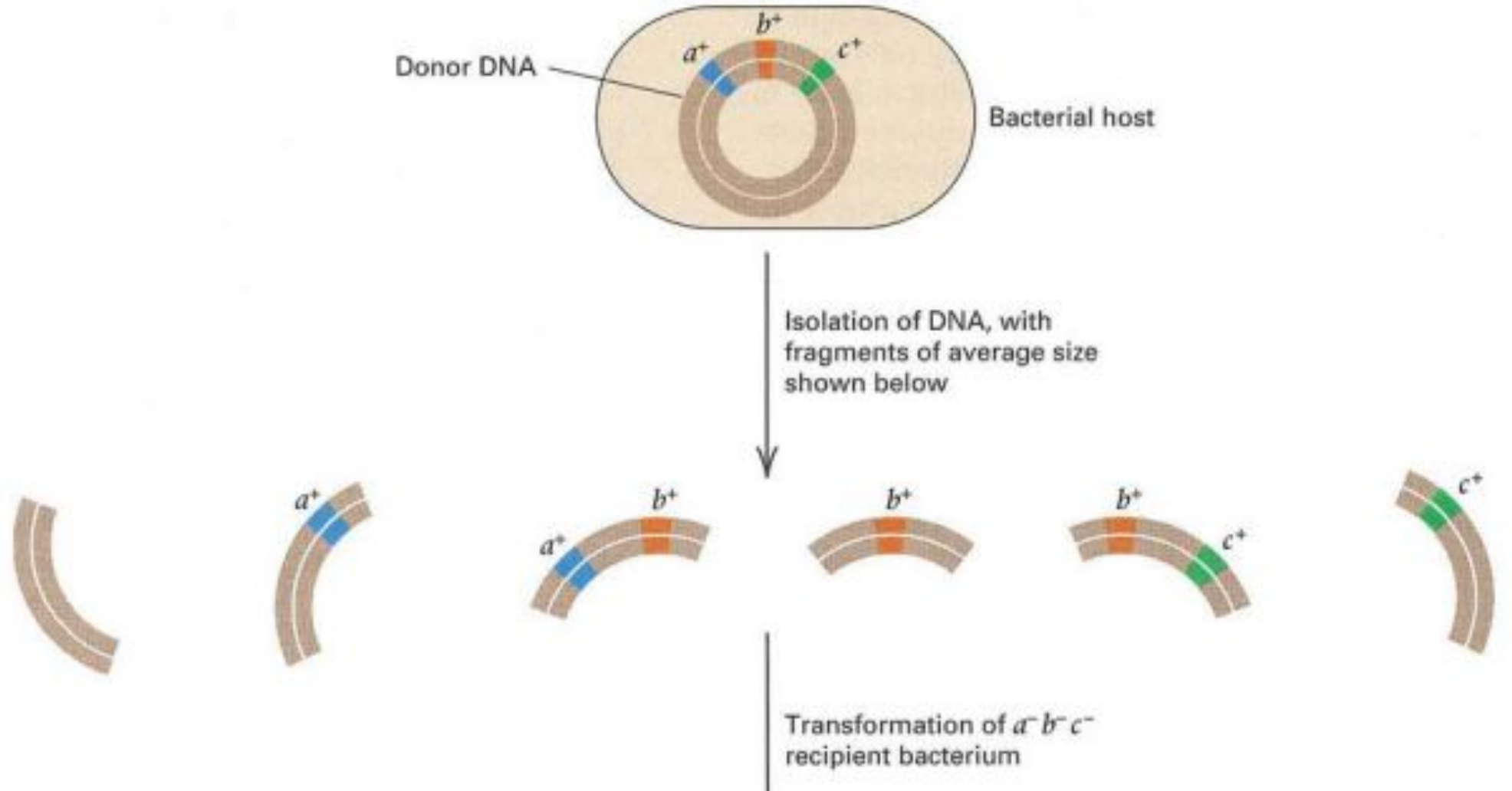
Transformation

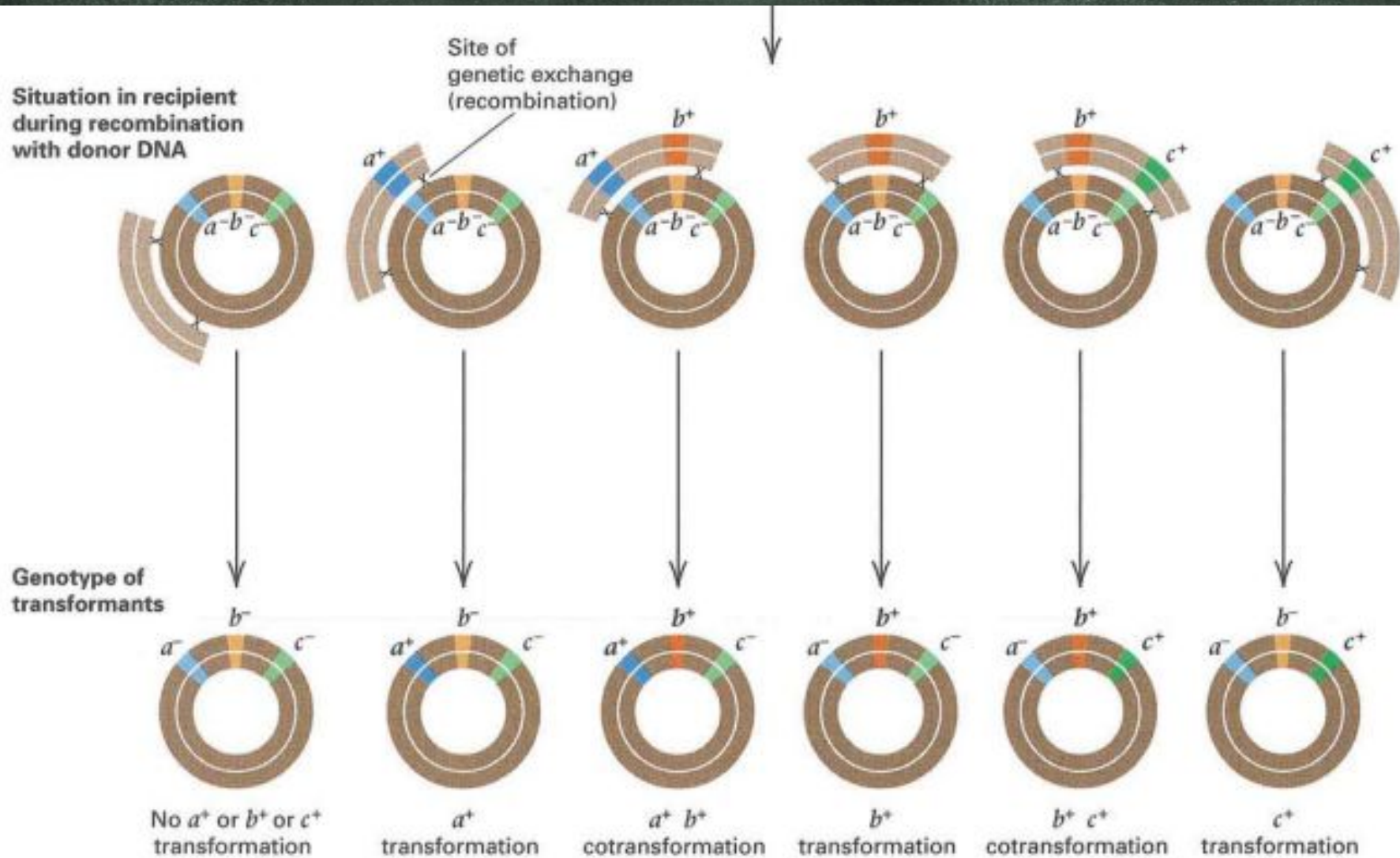


When a bacterial cell dies, it releases a fragment of its DNA into the environment. This DNA fragment is taken up by a recipient cell, which incorporates the DNA into its chromosome.



Transformation





No $a^+ b^+ c^+$ or $a^+ b^- c^+$ cotransformation occurs because the distance between a and c is too great.



Conjugation and Recombination

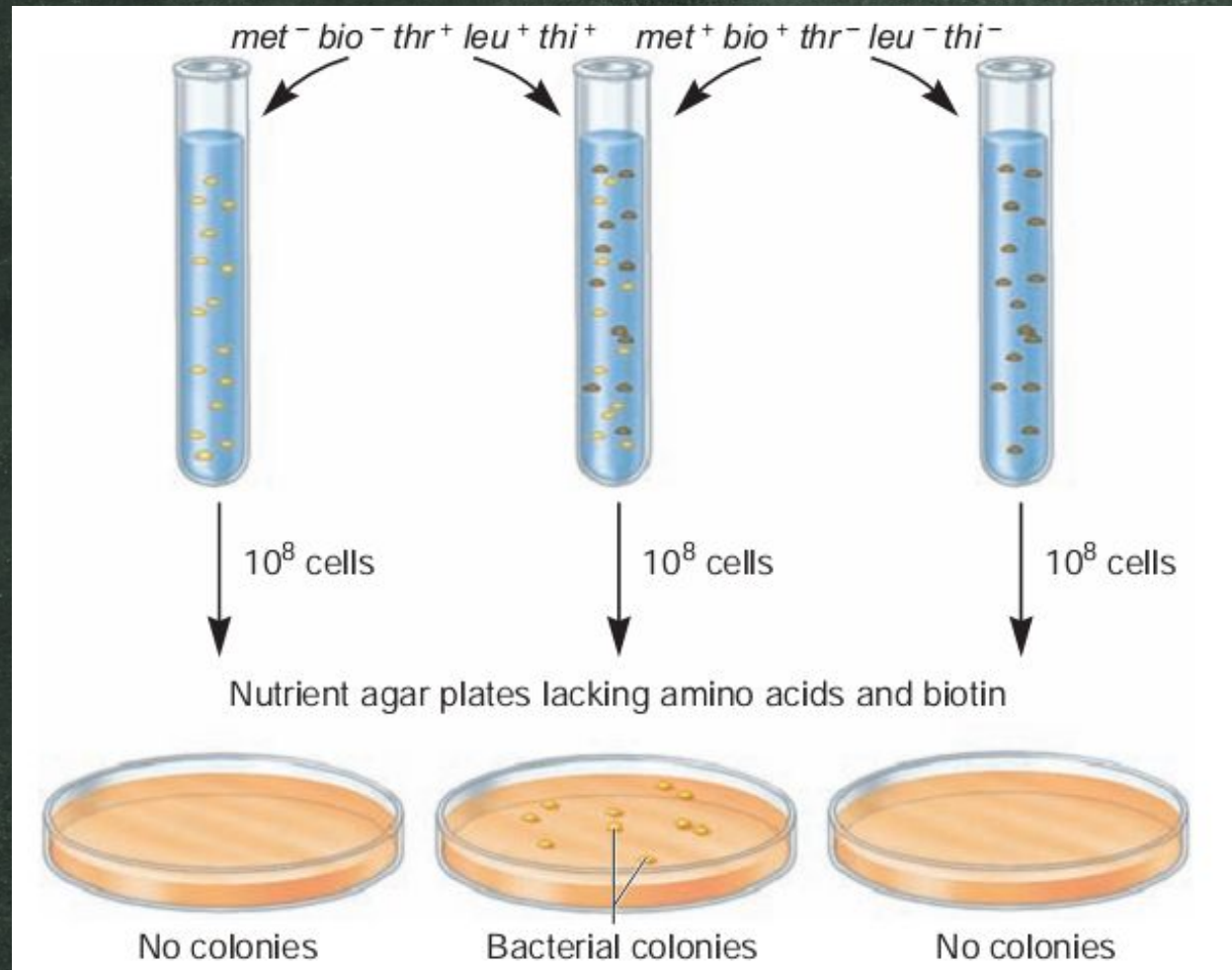
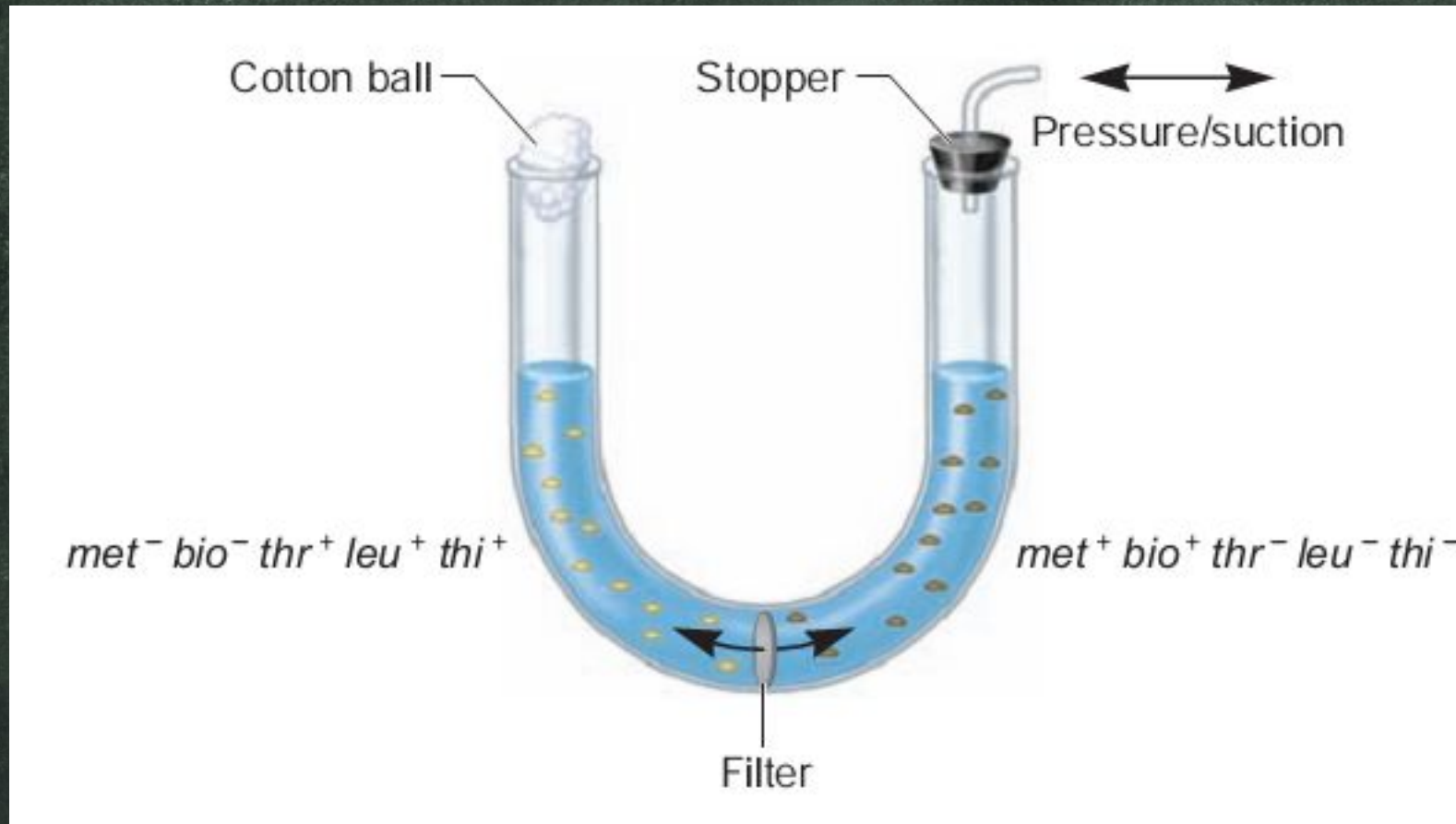


FIGURE 6.1 Experiment of Lederberg and Tatum demonstrating genetic transfer during conjugation in *E. coli*. When

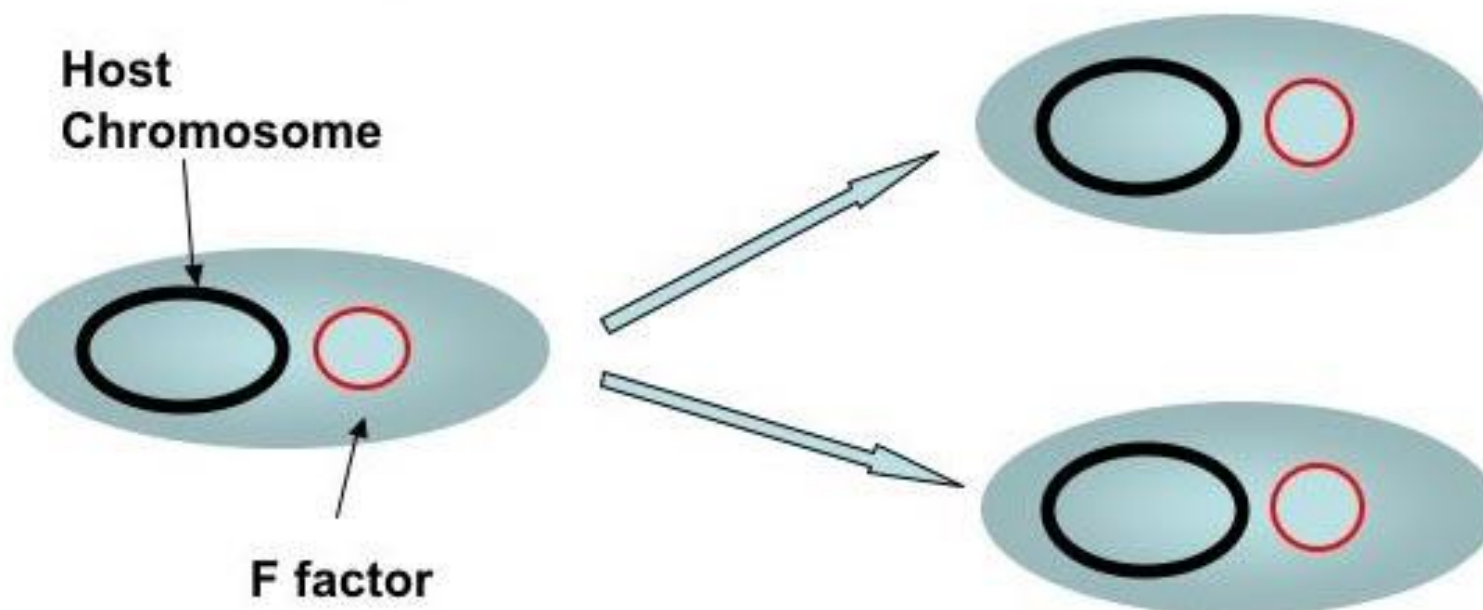


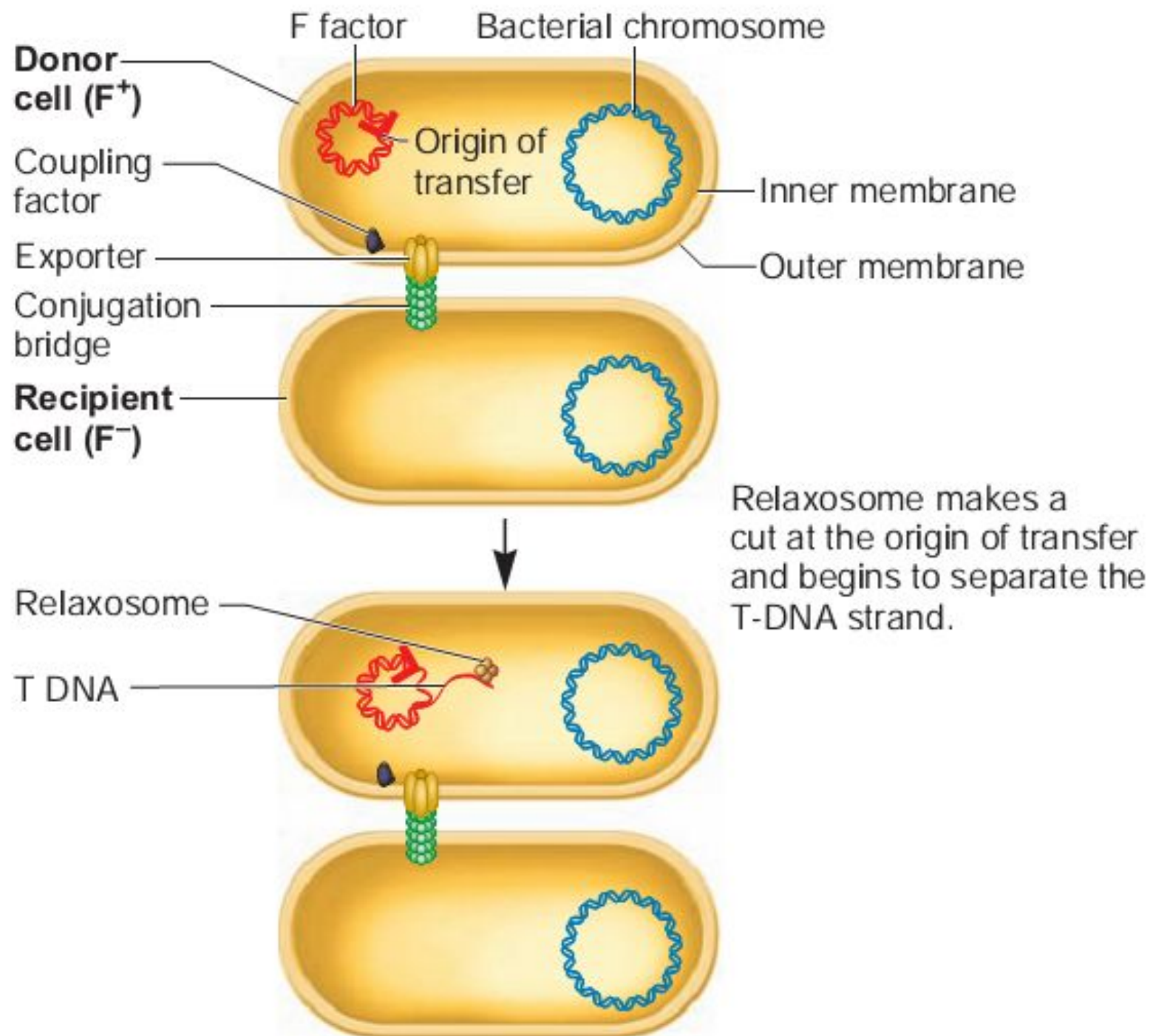
Conjugation needs direct cell-to-cell contact

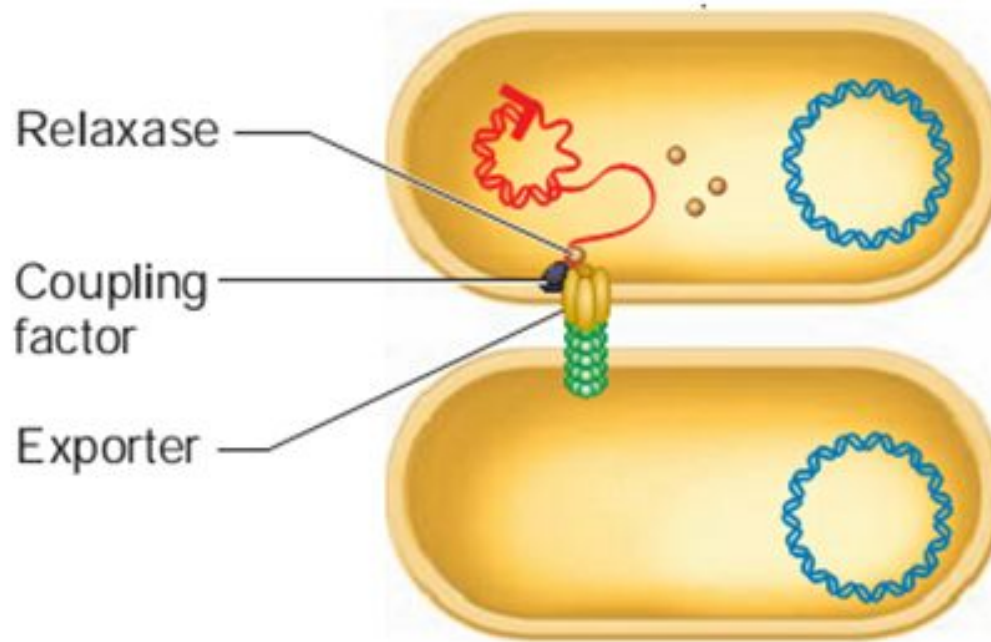


Conjugation and Recombination

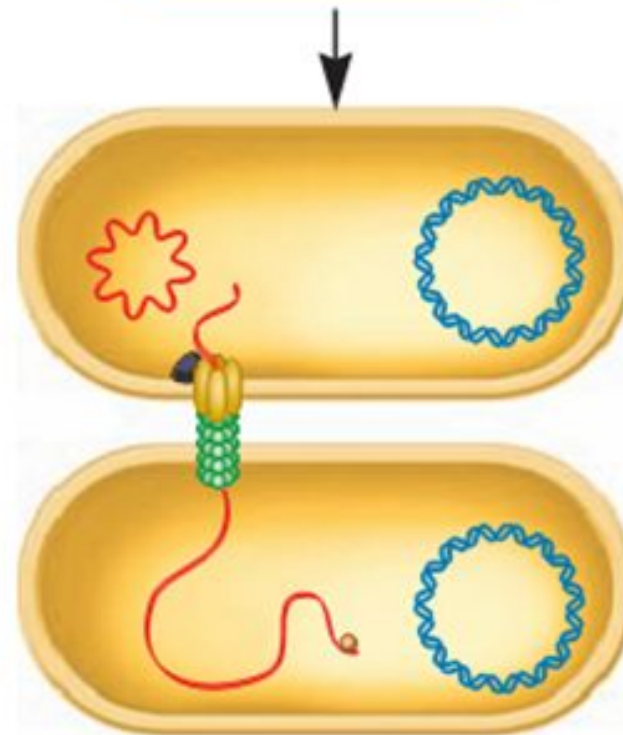
F replicates in cytoplasm and passed to daughter cells







Most proteins of the relaxosome are released. The T-DNA/relaxase complex is recognized by the coupling factor and transferred to the exporter.



The exporter pumps the T-DNA/relaxase complex into the recipient cell.



F⁺ cell



F⁺ cell

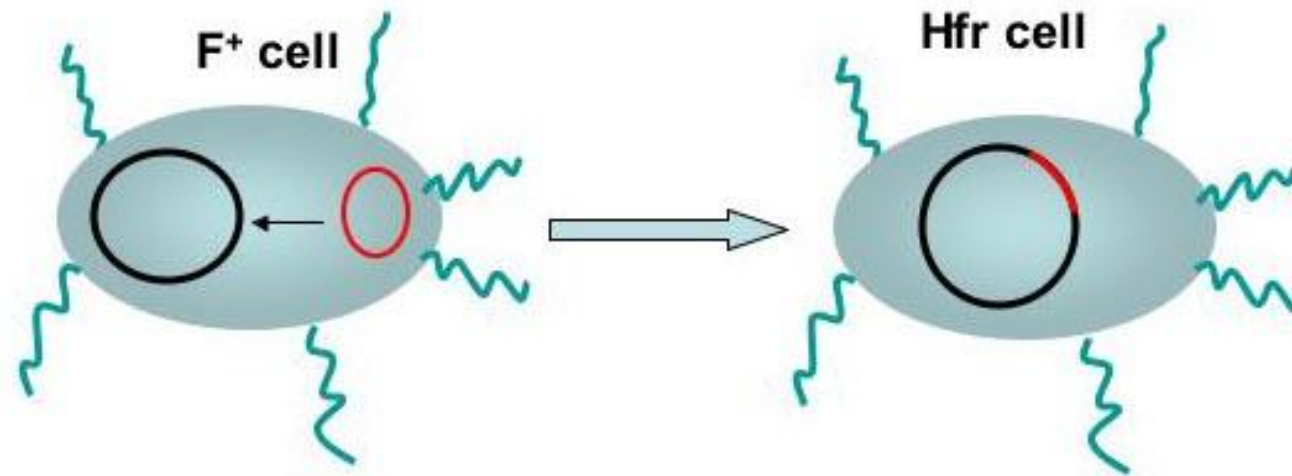


In the donor cell, the F-factor DNA is replicated to become double stranded. In the recipient cell, relaxase joins the ends of the T-DNA strand. It is then replicated to become double stranded.

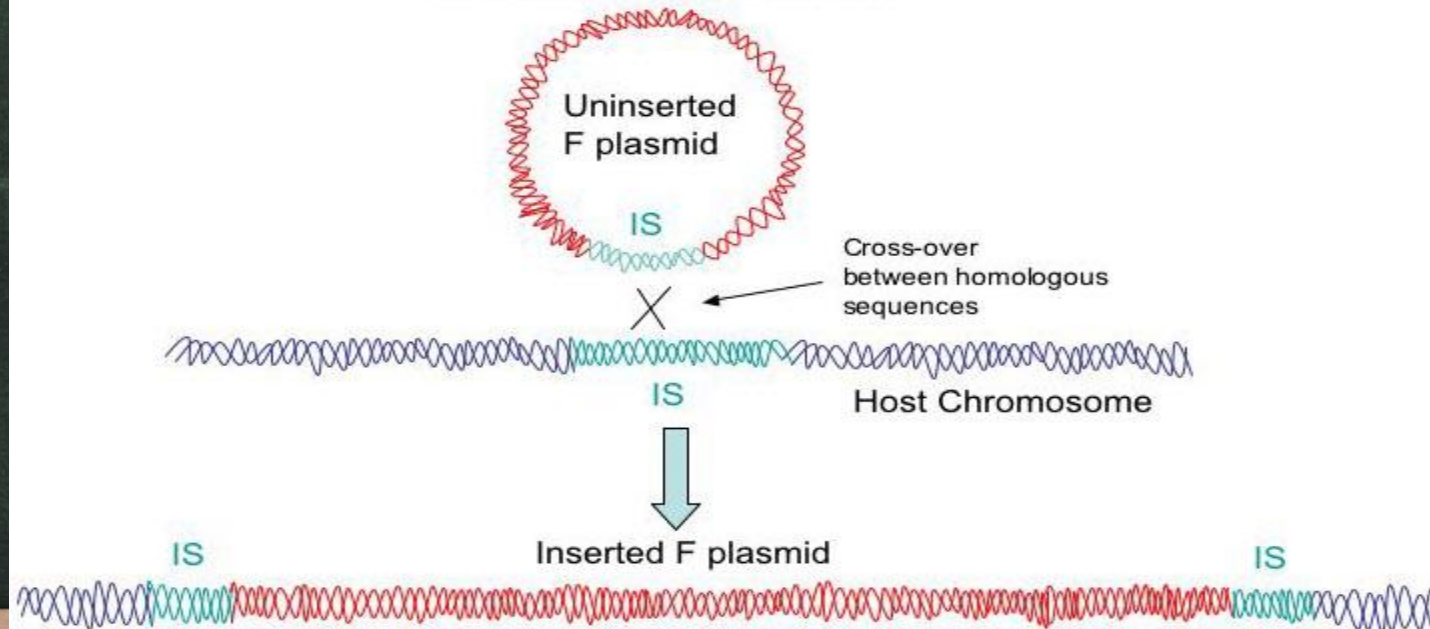
(a) Transfer of an F factor via conjugation

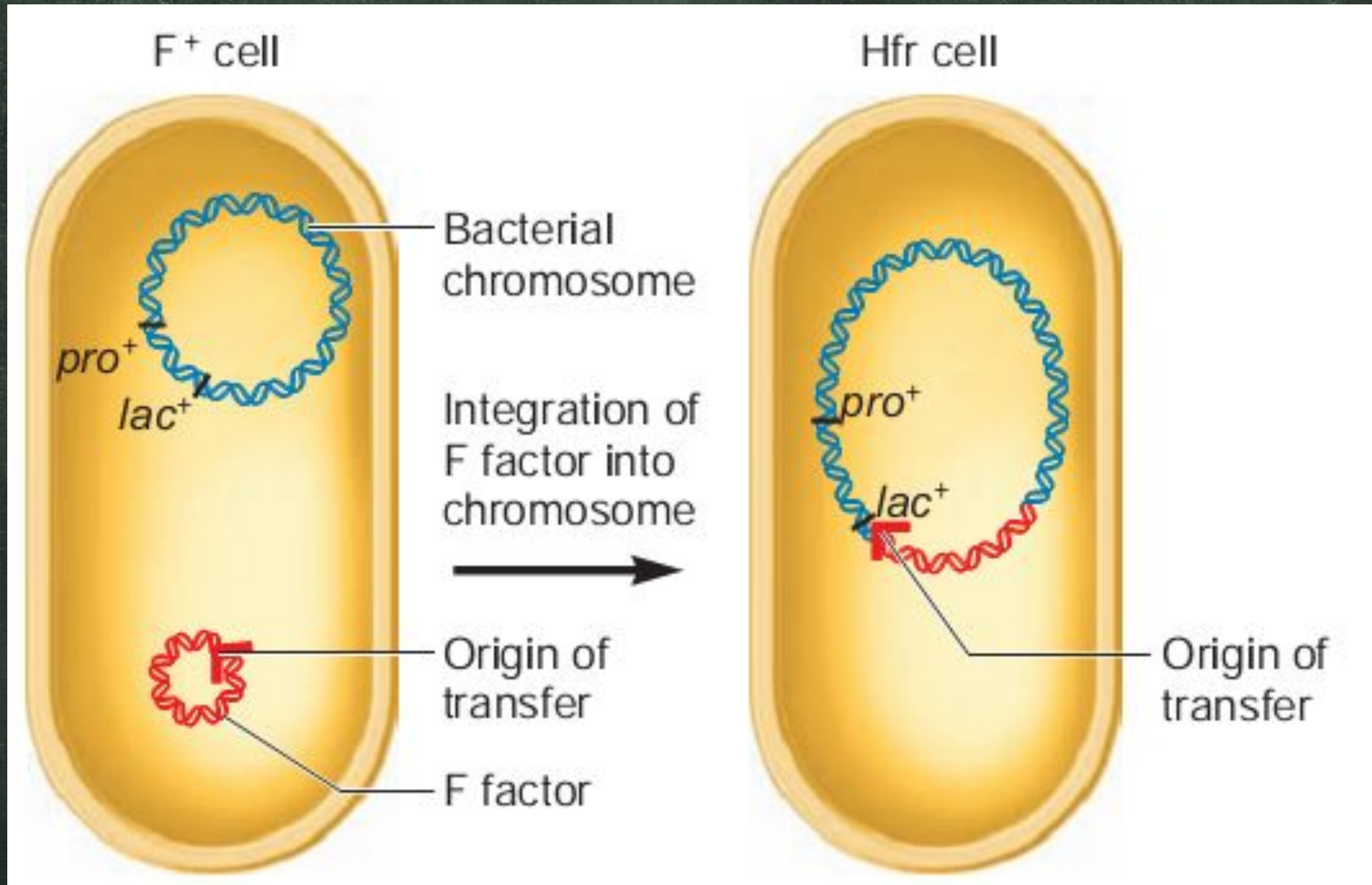


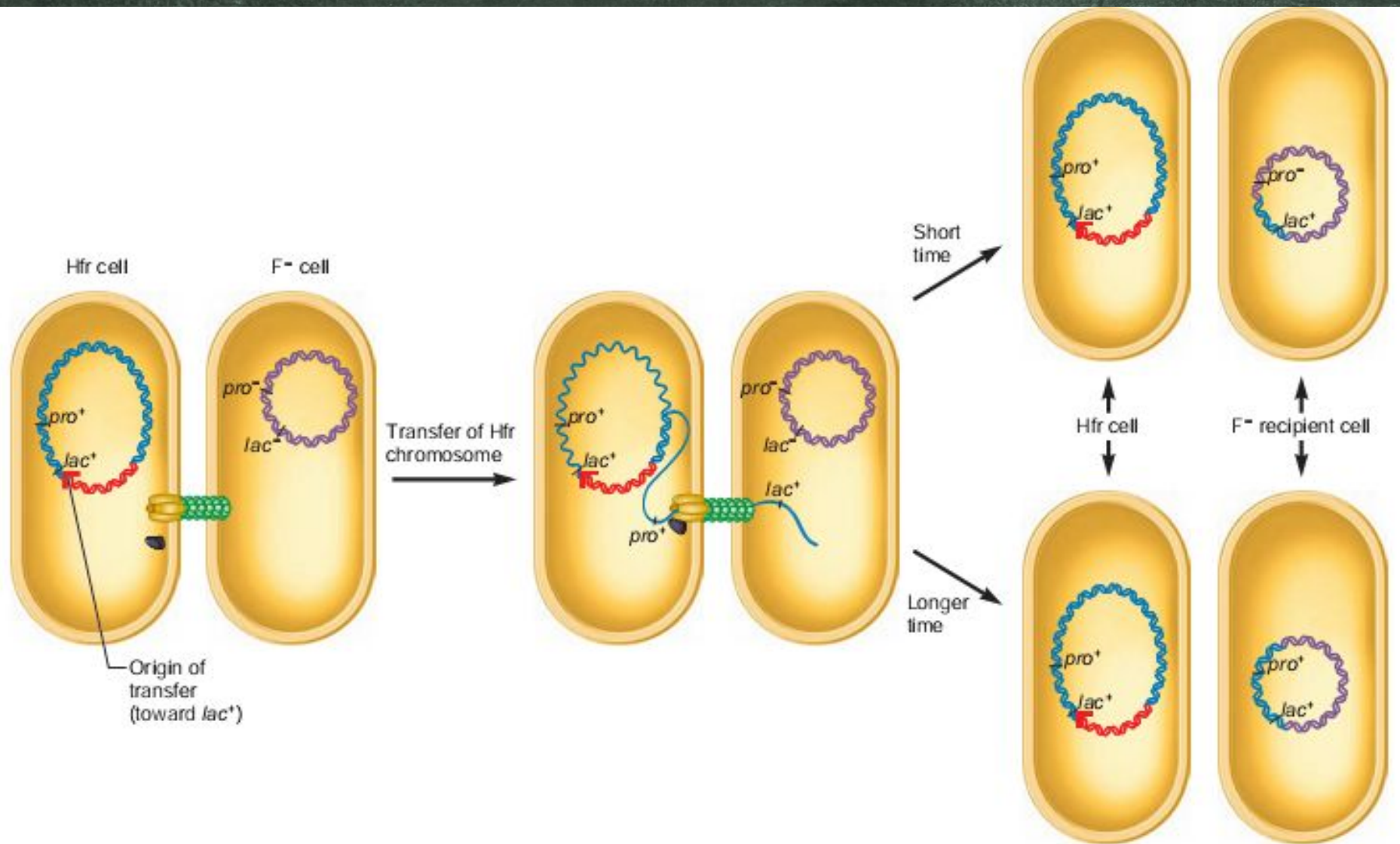
On rare occasions, F integrates into host chromosome



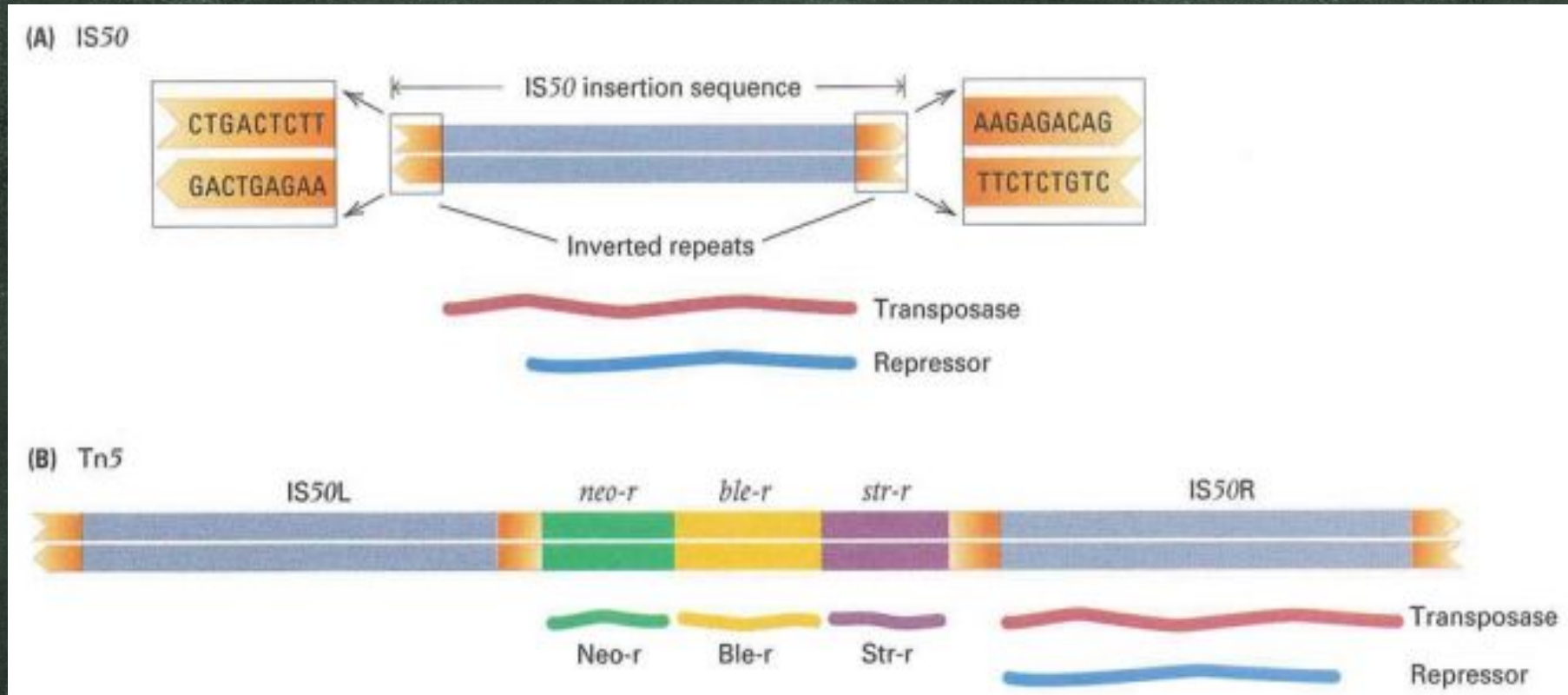
Integration of F plasmid



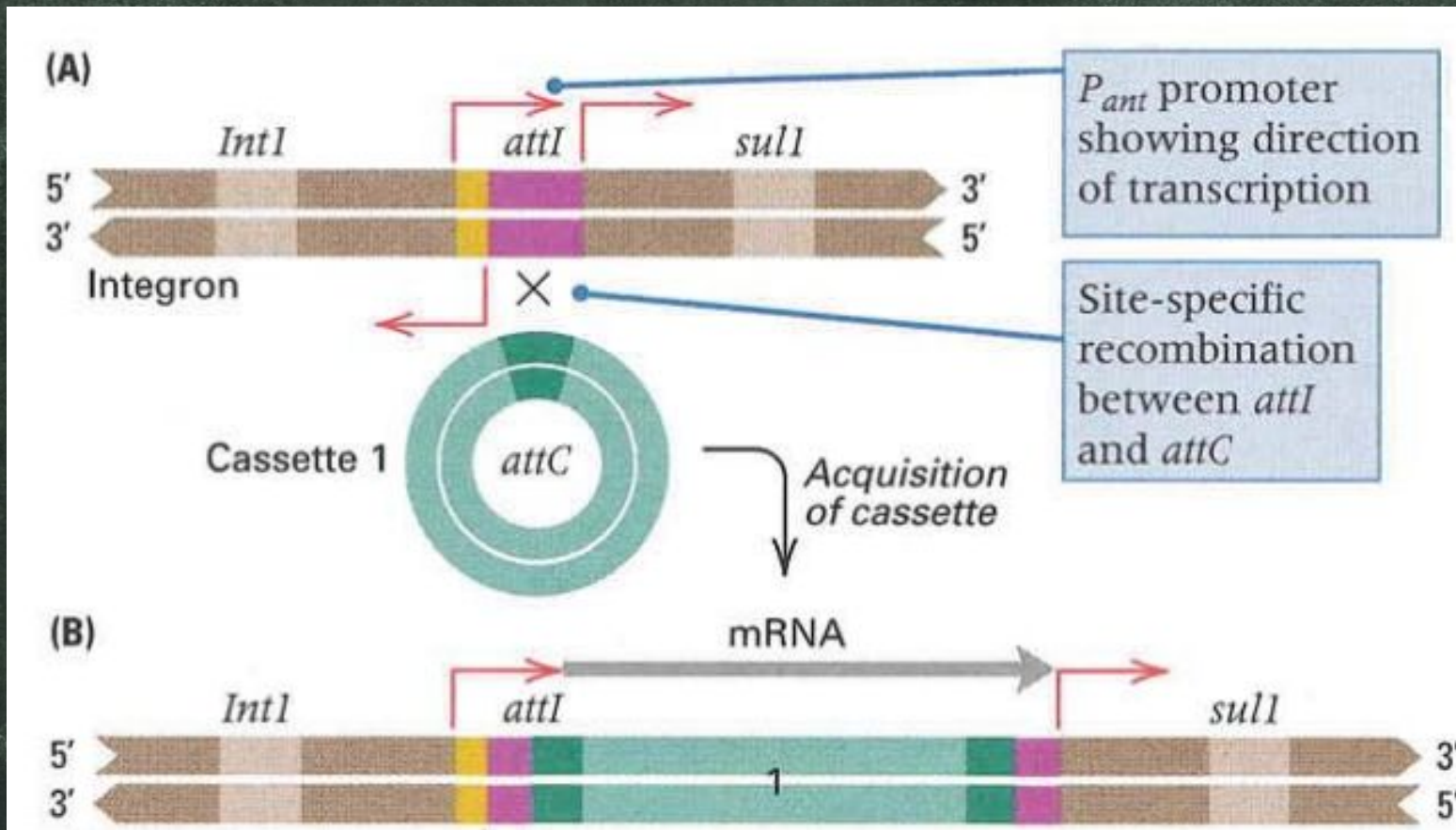




(b) An Hfr donor cell can pass a portion of its chromosome to an F⁻ recipient cell.



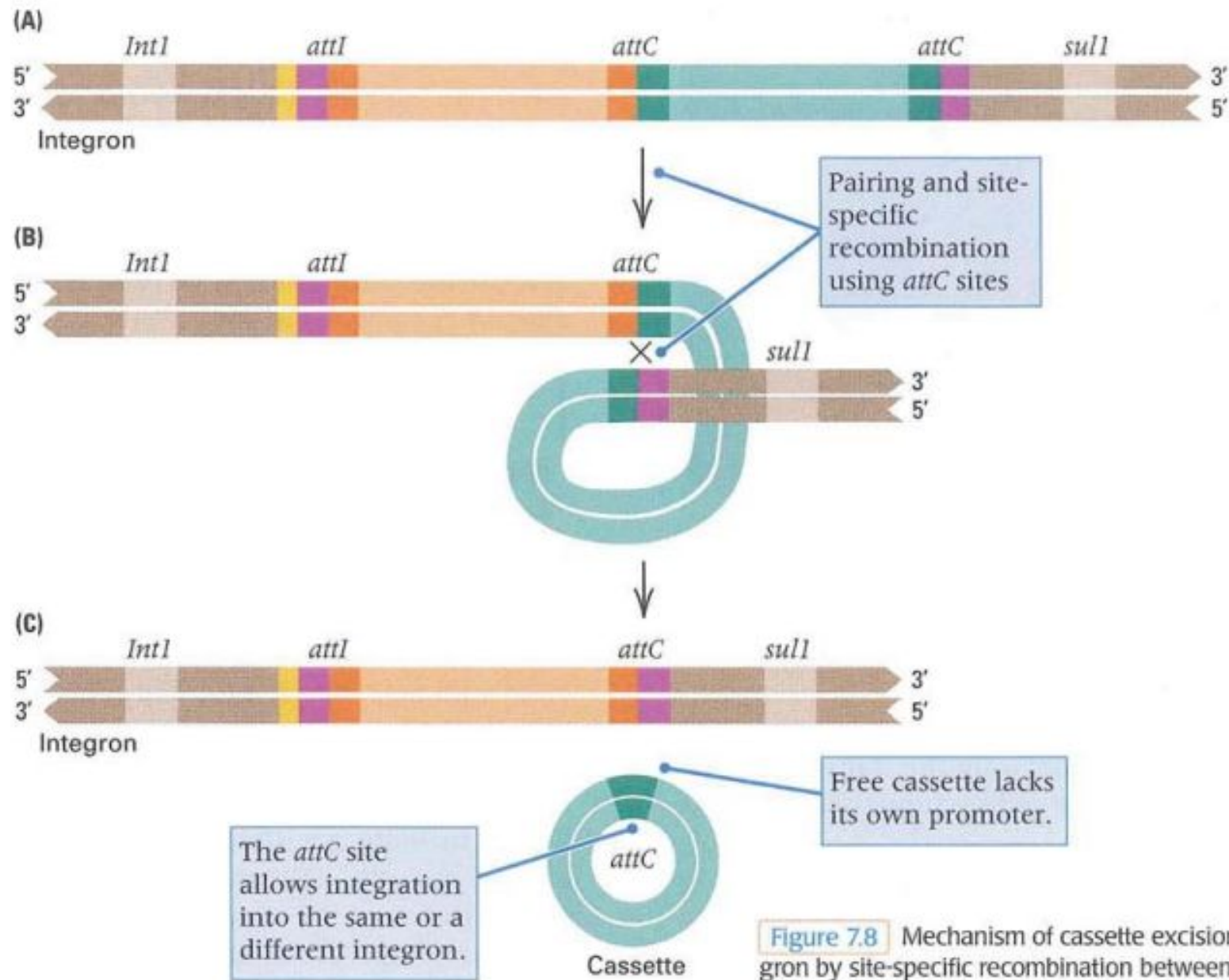
Insertion sequences and transposons play a key role in bacterial populations

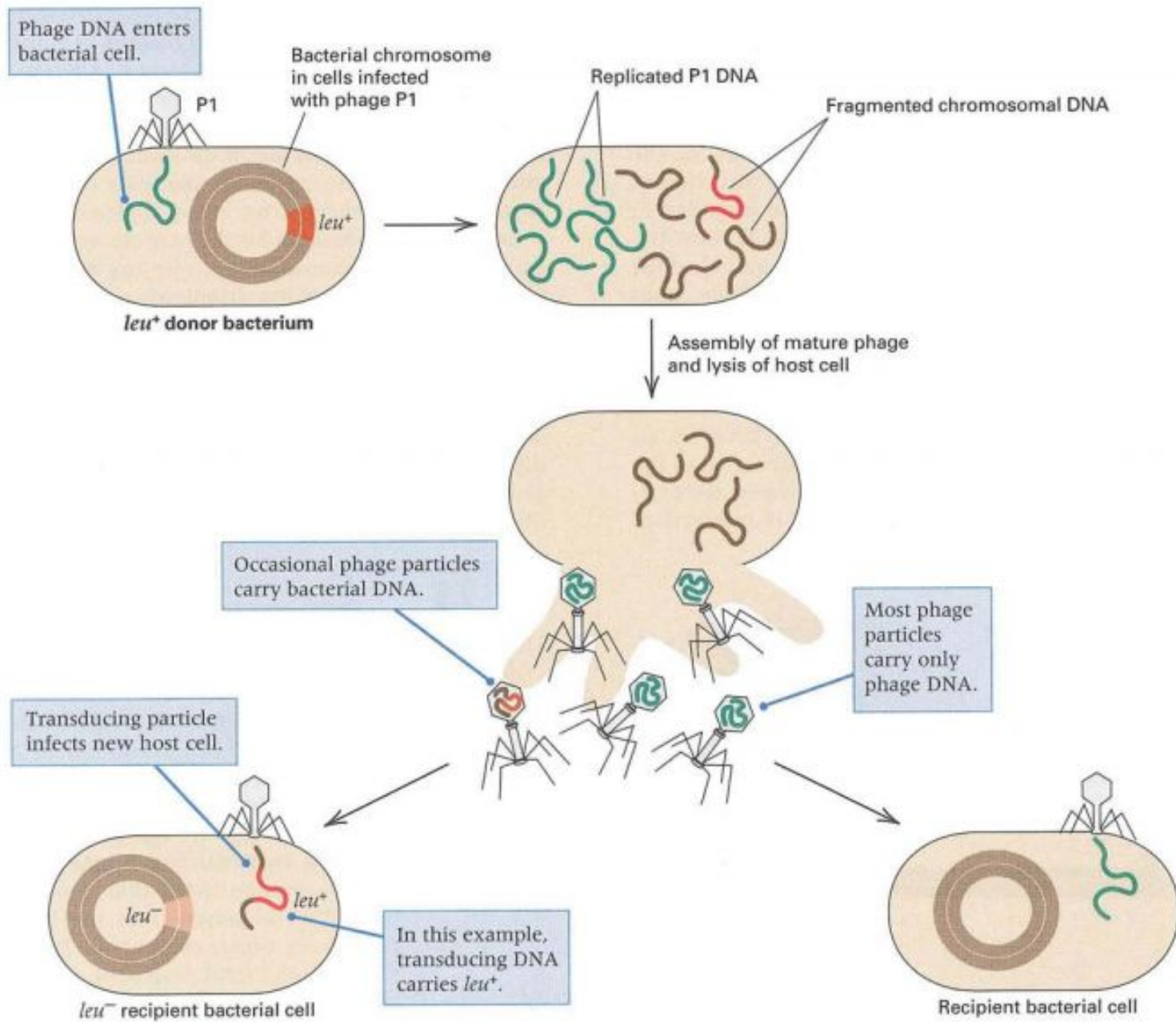


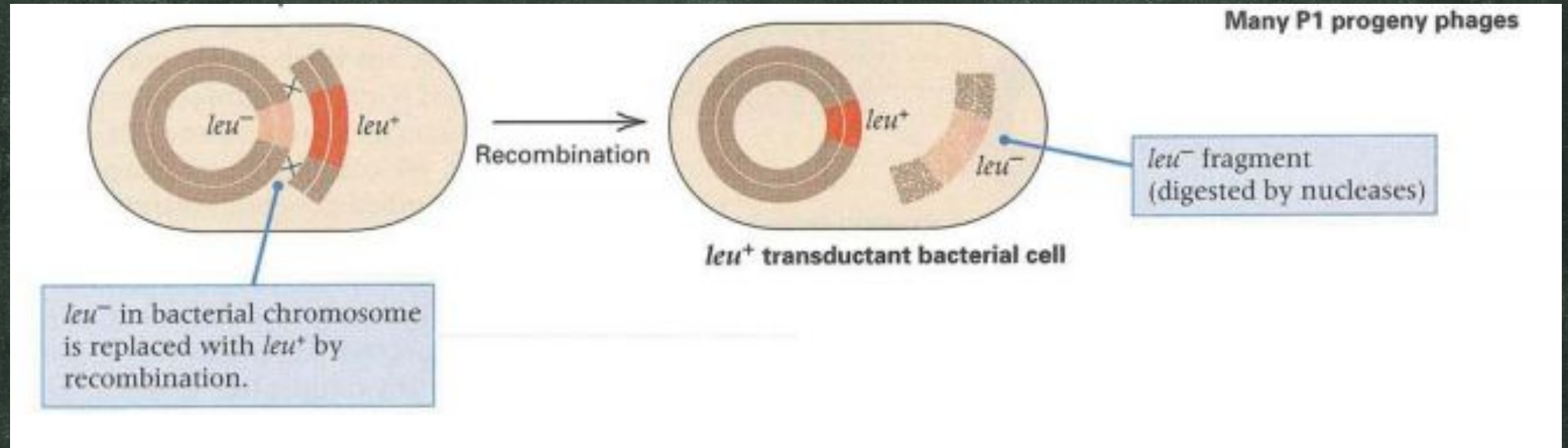
Gene capture and expression by integron

Integron: a DNA element that encodes a site-specific recombinase as well as a recognition region that allows other sequences with similar recognition regions to be incorporated into the integron by recombination

Cassette: circular antibiotic-resistance-coding region flanked by a recognition region for an integron

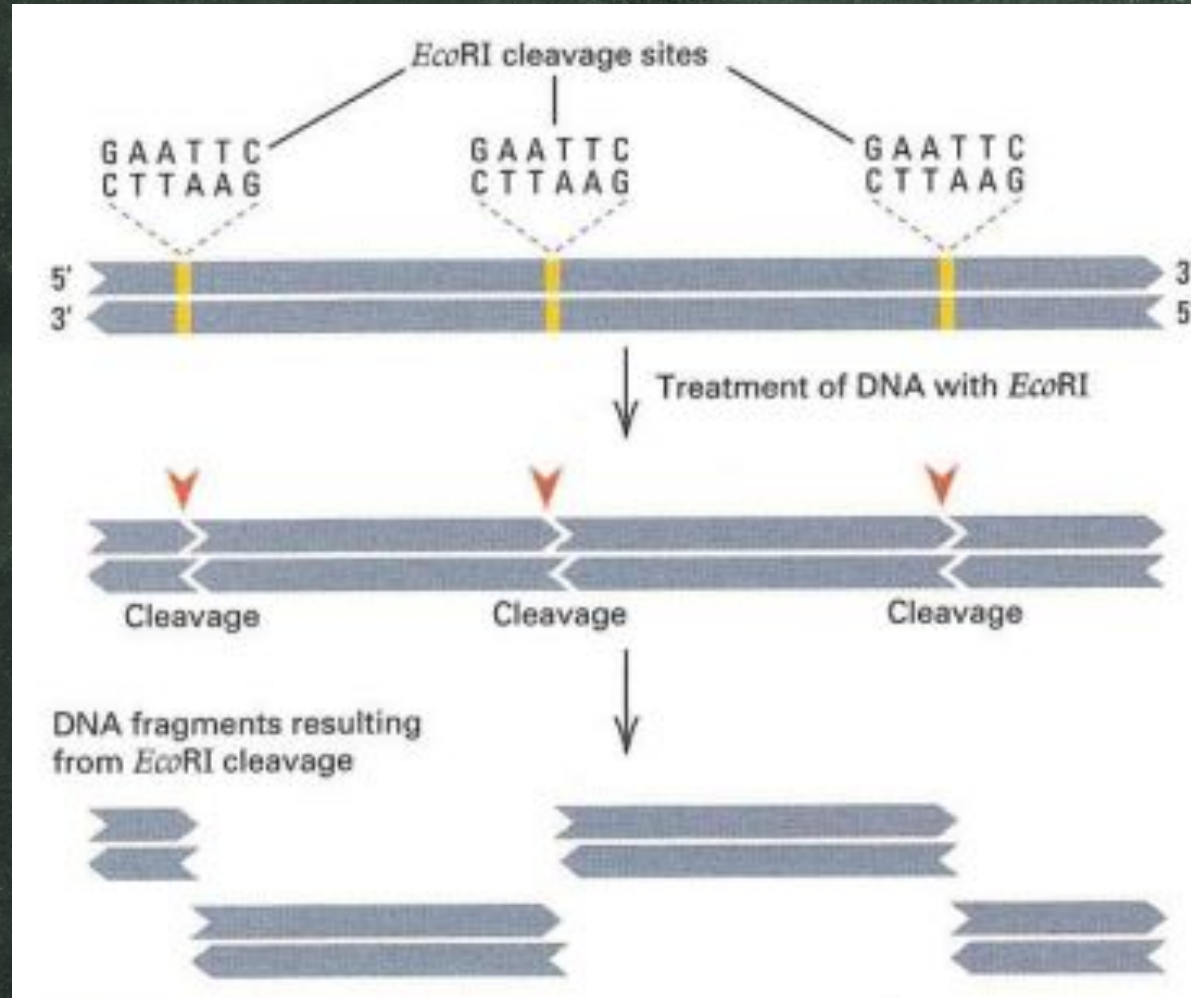








Use of restriction enzyme for mapping





Restriction Map Practice Question:

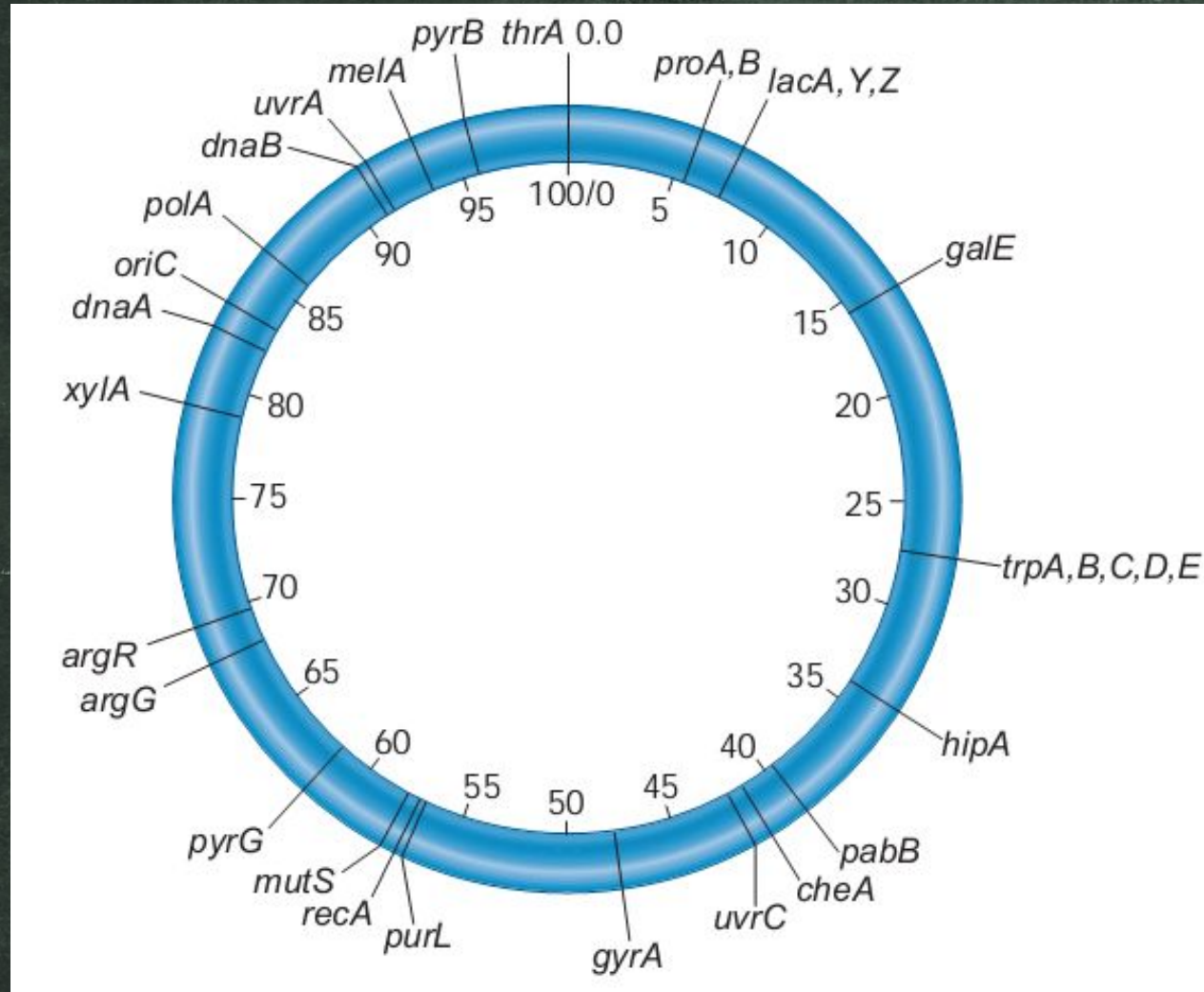
A linear fragment of DNA (7.5 kb) is cleaved with the individual restriction enzymes *HindIII* and *SmaI* and then with a combination of the two enzymes. The fragments obtained are:

<i>HindIII</i> :	2.5 kb, 5 kb
<i>SmaI</i> :	2.0 kb, 5.5 kbs
<i>HindIII</i> and <i>SmaI</i> :	2.5 kb, 3.0 kb, 2.0 kb.

- a. Draw the restriction map.

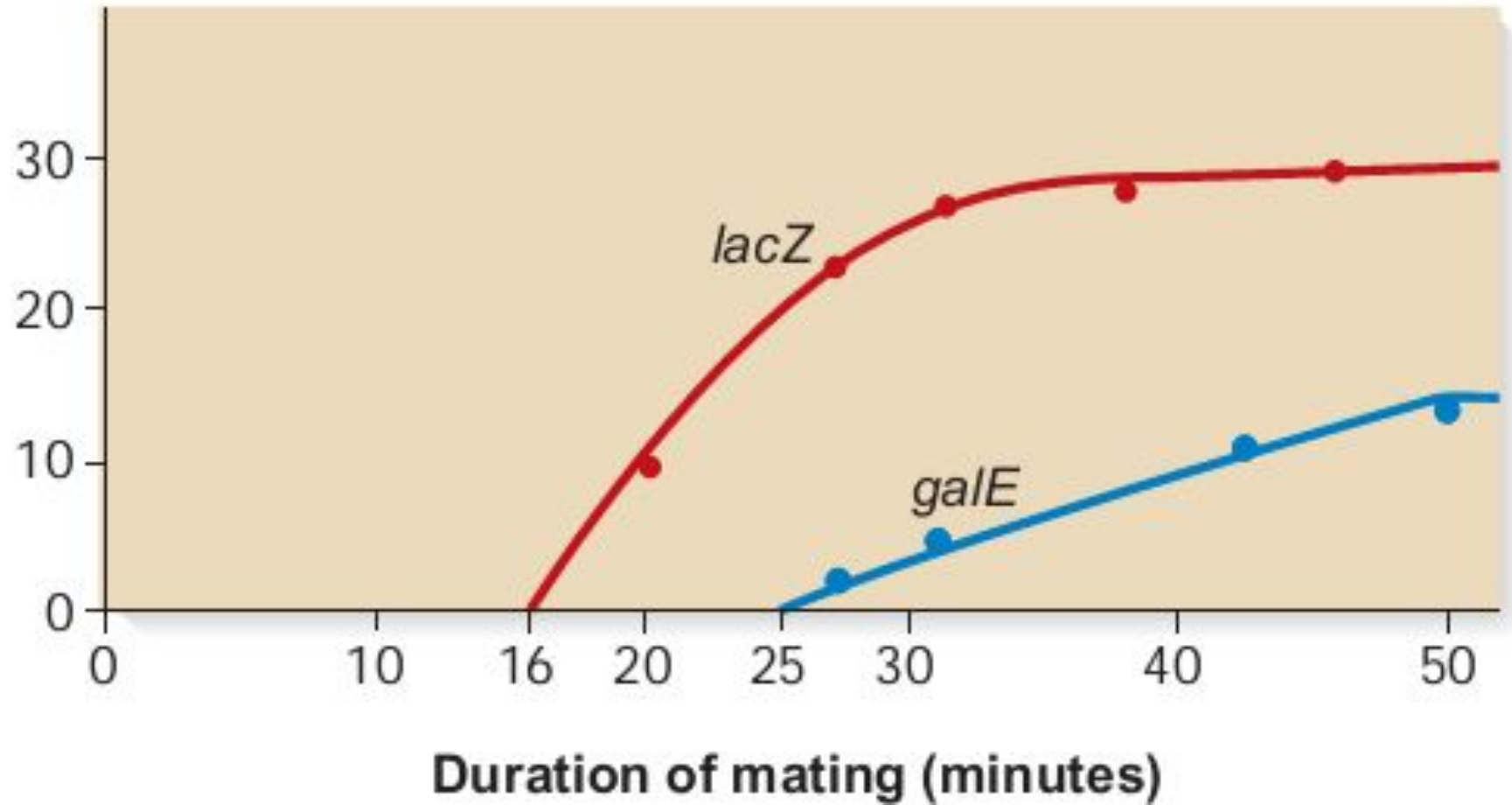


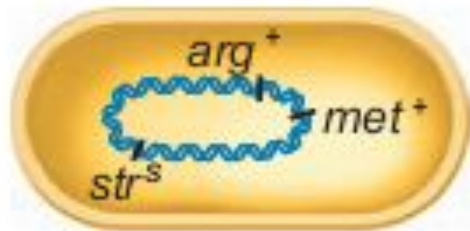
Simplified genetic map of the *E.coli*





% of F⁻ recipient cells
that have received the
gene during conjugation



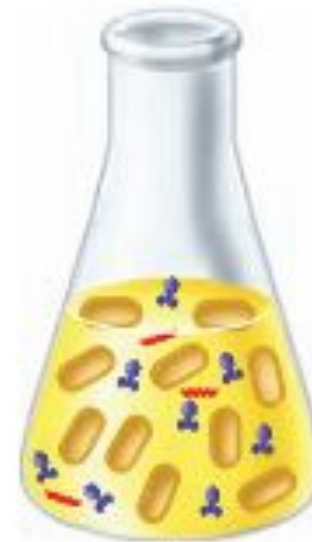


Donor cell

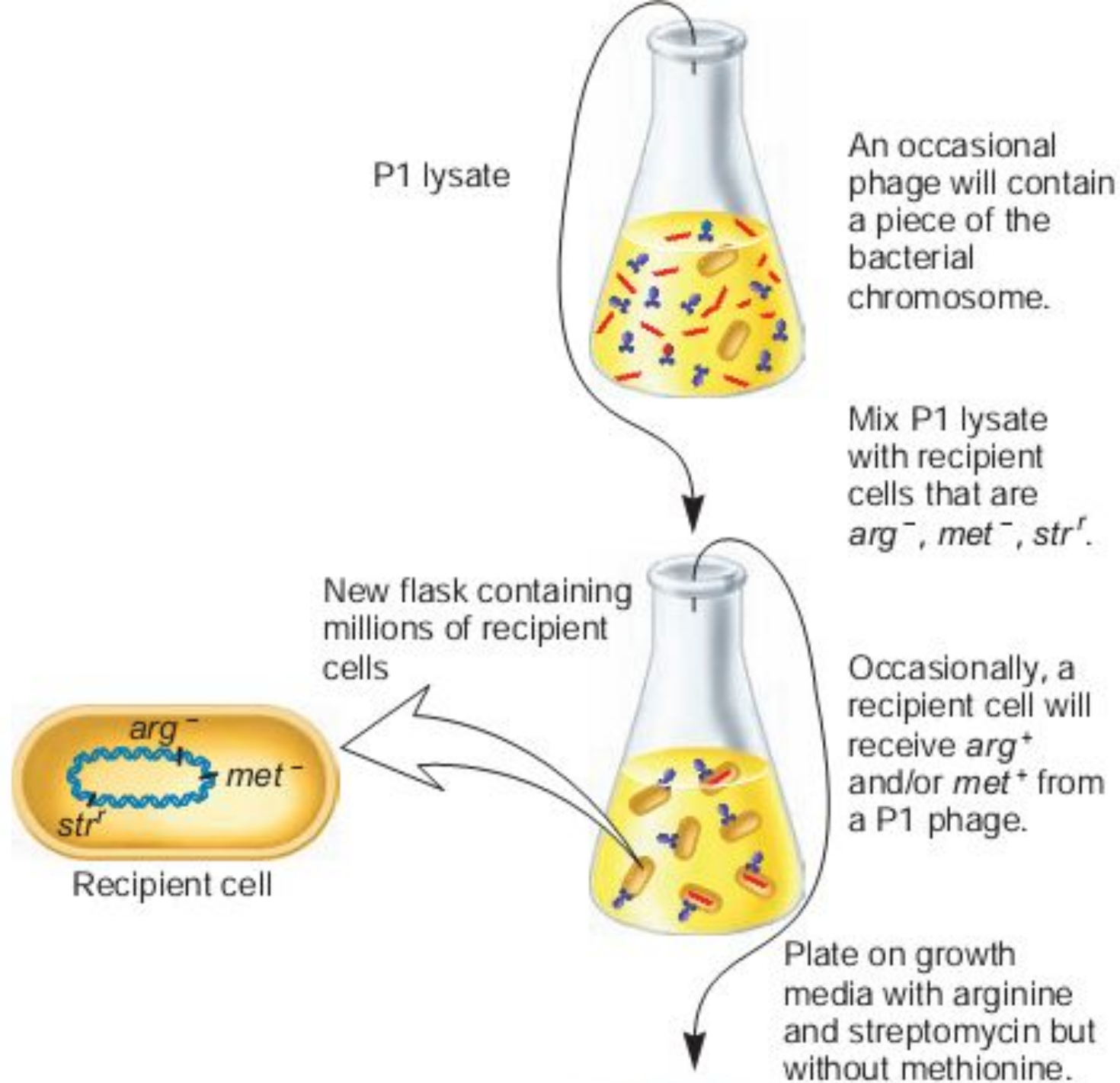


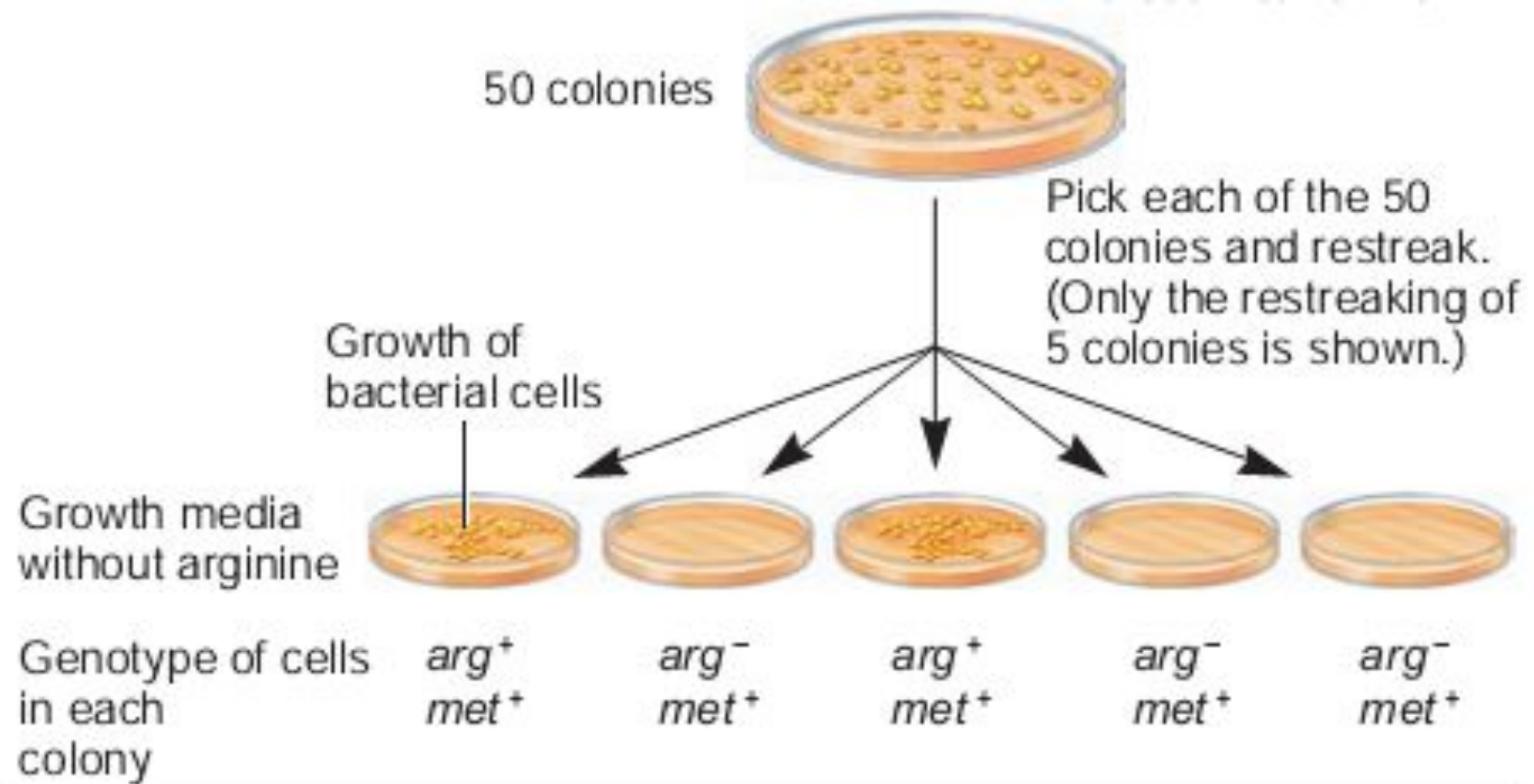
P1

Donor strain: $\text{arg}^+ \text{met}^+ \text{str}^s$
Recipient strain: $\text{arg}^- \text{met}^- \text{str}^r$



Infection, production of
new phages and lysis





Results				
Selected gene	Nonselected gene	Number of colonies that grew on		Cotransduction frequency
		media + arginine	media - arginine	
met^+	arg^+	50	21	0.42



$$\text{Cotransduction frequency} = (1 - d/L)^3$$

where

d = distance between two genes in minutes

L = the size of the chromosomal pieces (in minutes) that the phage carries during transduction (For P1 transduction, this size is approximately 2% of the bacterial chromosome, which equals about 2 minutes.)

$$0.42 = (1 - d/2)^3$$

$$(1 - d/2) = \sqrt[3]{0.42}$$

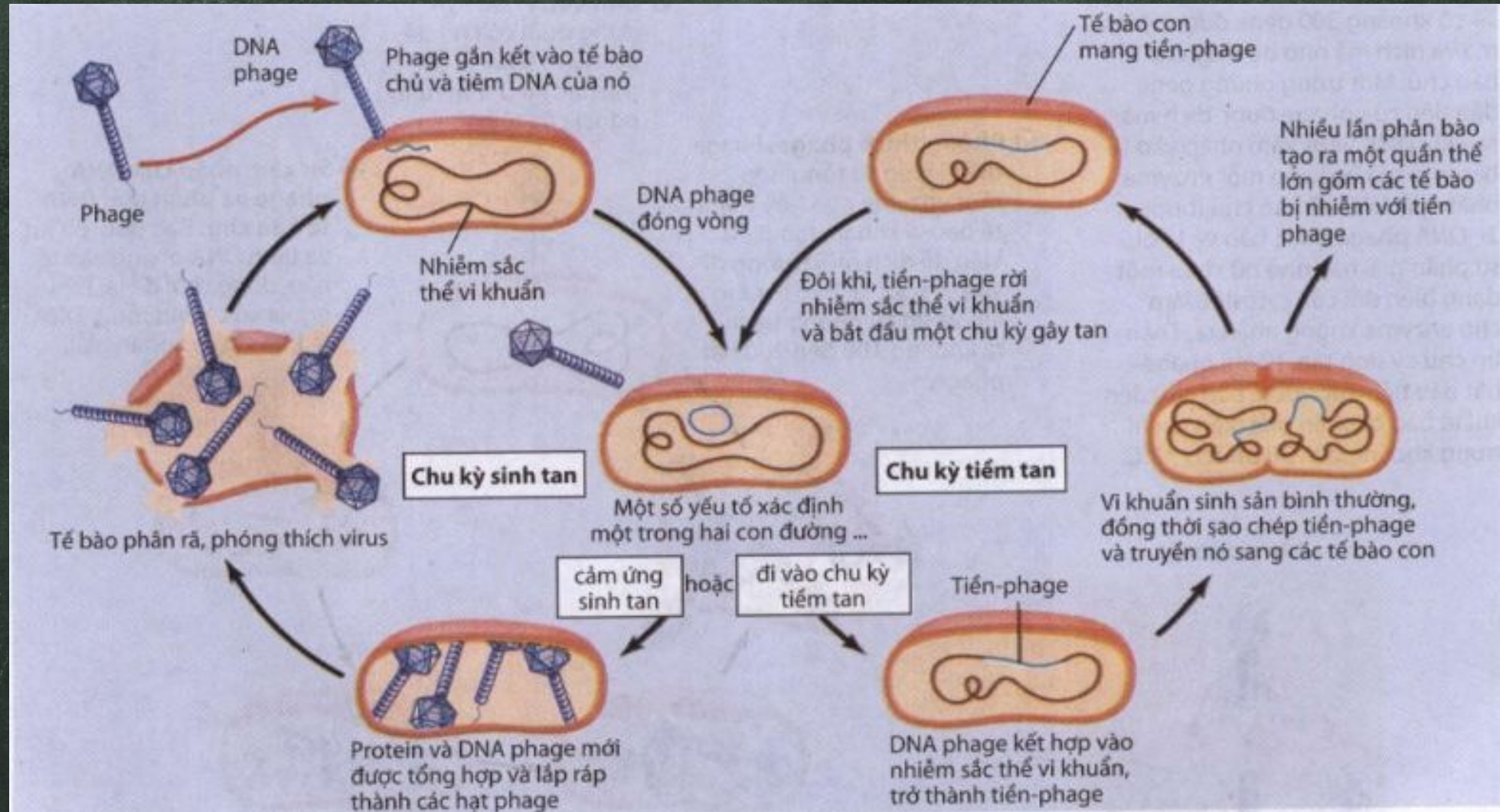
$$1 - d/2 = 0.75$$

$$d/2 = 0.25$$

$$d = 0.5 \text{ minutes}$$

This equation tells us that the distance between the *met*⁺ and *arg*⁺ genes is approximately 0.5 minutes.

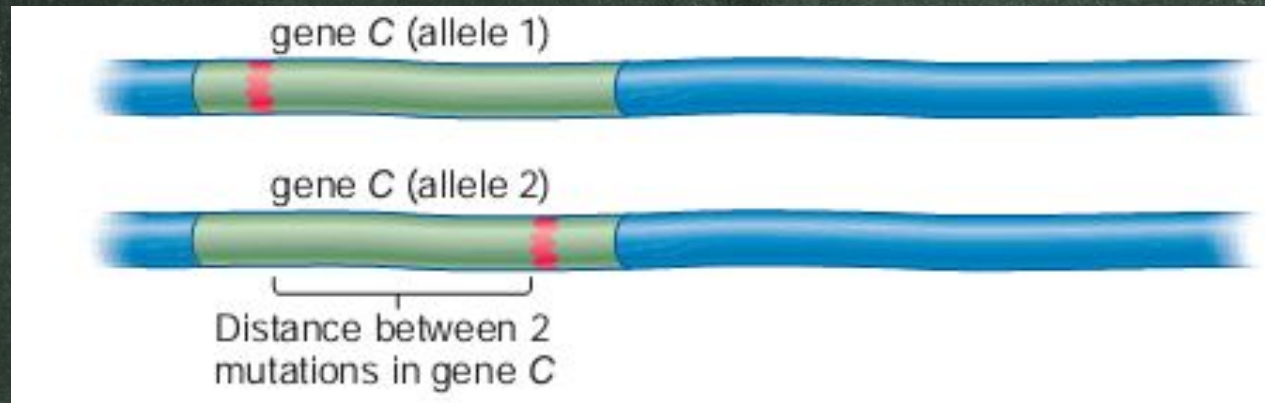
Chapter 11: Viral Genetics



Lytic-lysogenic cycle in viruses



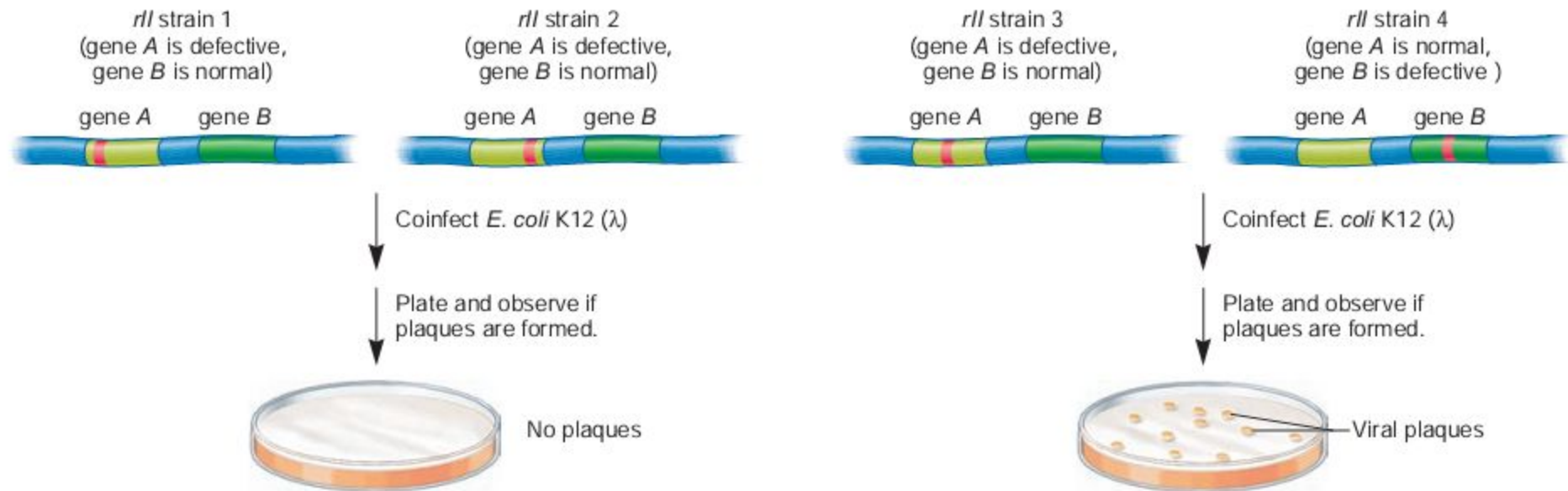
Intragenic mapping (fine structure mapping) in bacteriophage



Seek to establish distances between 2 or more mutations within the same gene



A complement test can reveal if mutations are in the same gene or in different genes

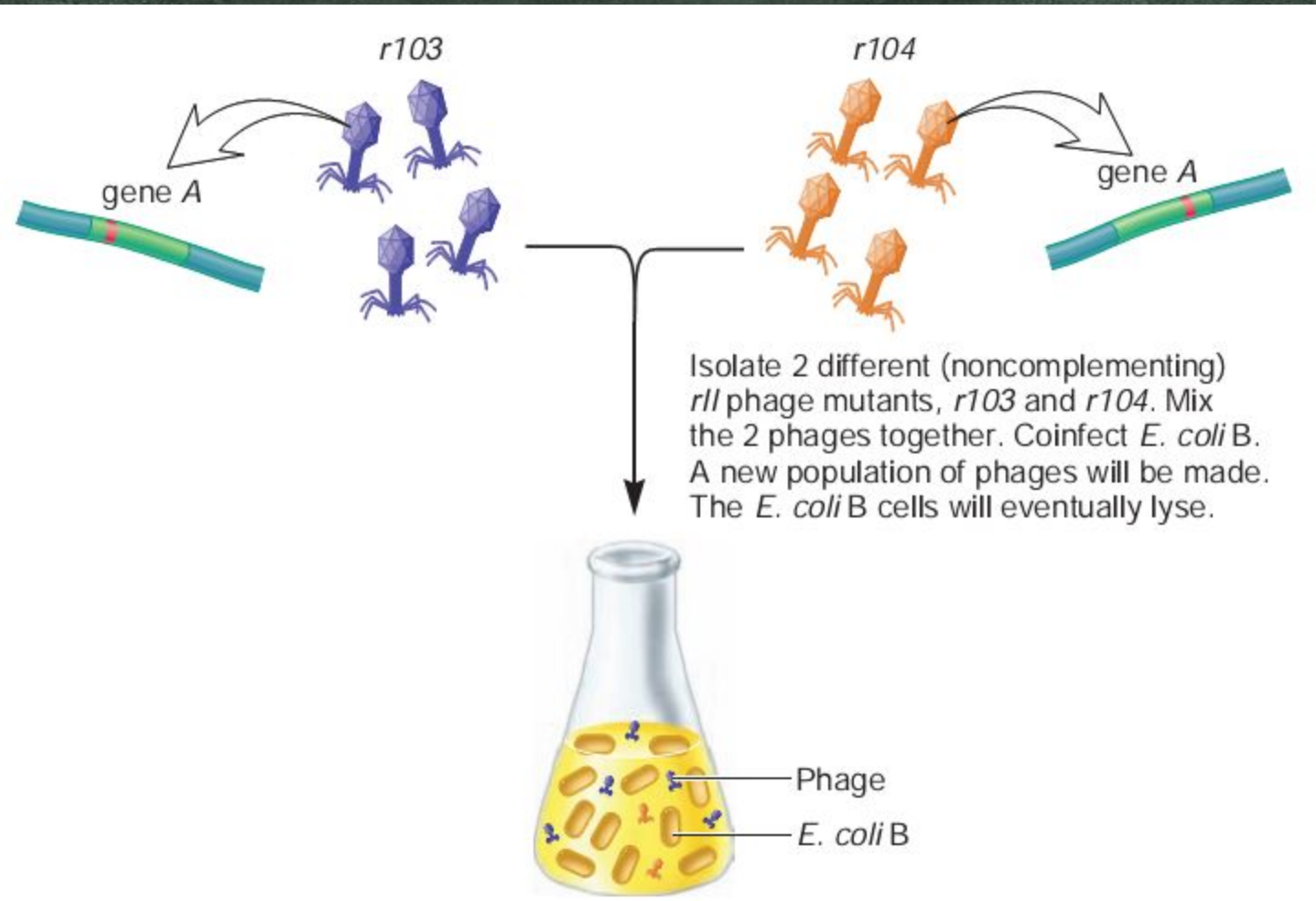


No complementation occurs, because the coinfecting cell is unable to make the normal product of gene A. The coinfecting cell will not produce viral particles, thus no bacterial cell lysis and no plaque formation.

(a) Noncomplementation: The phage mutations are in the same gene.

Complementation occurs, because the coinfecting cell is able to make normal products of gene A and gene B. The coinfecting bacterial cell will produce viral particles that lyse the cell, resulting in the appearance of clear plaques.

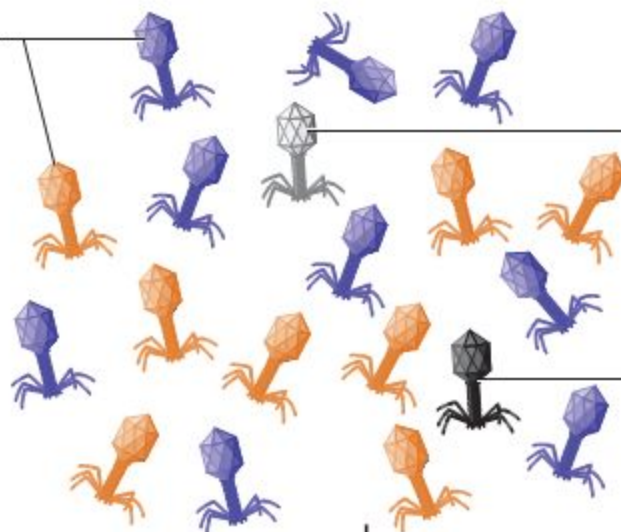
(b) Complementation: The phage mutations are in different genes.





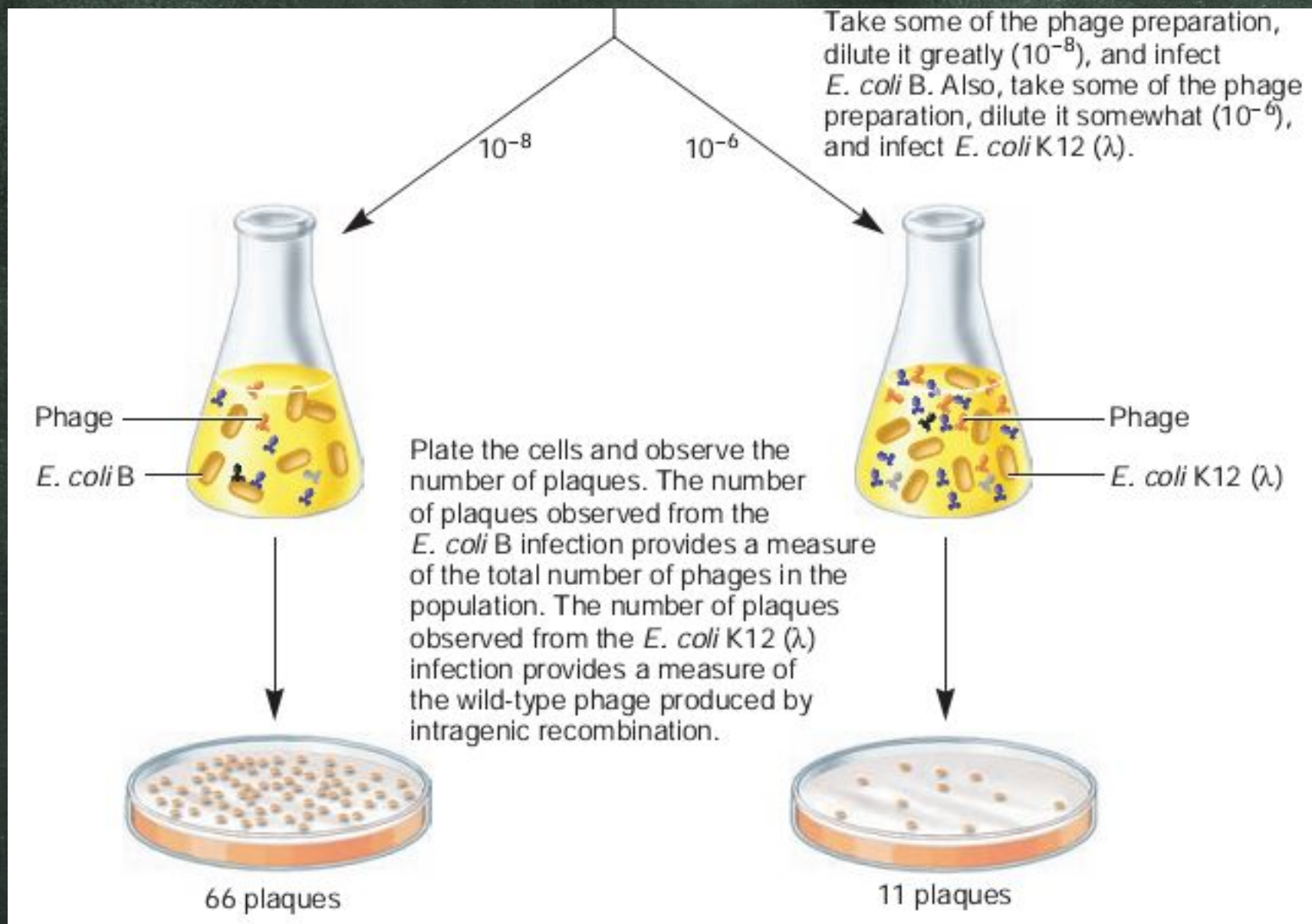
Isolate this new population of phages. It will primarily contain nonrecombinant phages, but it will occasionally contain intragenic recombinants of wild type and double mutant phages (depicted in white and black, respectively). The phage preparation can contain several billion phages per milliliter.

Nonrecombinant
phages



Wild-type
phage

Double
mutant
phage

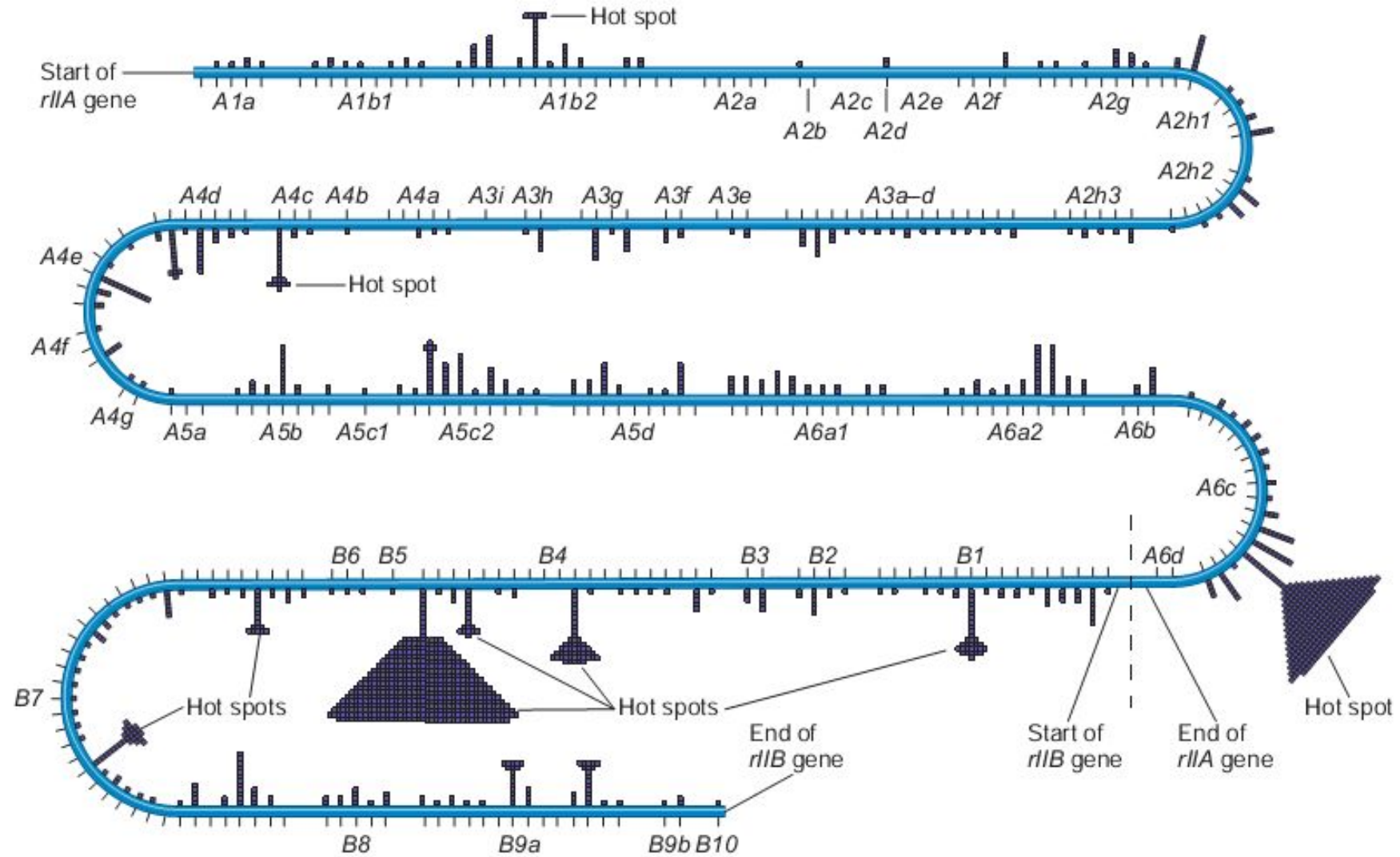




$$\text{Frequency of recombinants} = \frac{2 [\text{Wild-type plaques obtained in } E. coli \text{ K12}(\lambda)]}{\text{Total number of plaques obtained in } E. coli \text{ B}}$$

We can use this equation to calculate the frequency of recombinants obtained in the experiment described in Figure 6.18.

$$\begin{aligned}\text{Frequency of recombinants} &= \frac{2(11 \times 10^6)}{6.6 \times 10^9} \\ &= 3.3 \times 10^{-3} = 0.0033\end{aligned}$$





Population genetics

- Inheritance in Self-fertilizing Populations
 - Genotype frequencies at generation n :
 $(2n-1)AA:2Aa:(2n-1)aa$
- Inheritance in Random Mating Populations
 - Hardy-Weinberg Principle: $p^2AA:2pqAa:q^2aa$

Determining Population Equilibrium:

- Calculating T : (e.g., $T=P \times RH^2$ or related equilibrium coefficients)
- Chi-square (χ^2) test